

Project Management 412

Assignment 3: Post-completion project review

Group K1

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Abstract

This report details the successful execution and outcomes of the Automated Smart Distribution Warehouse project at Takealot. The project aimed to significantly enhance operational efficiency, safety, and technological sophistication within the distribution centre through the integration of advanced robotics, artificial intelligence (AI), and Internet of Things (IoT) technologies. Key objectives included increasing daily order throughput, reducing warehouse safety incidents, and automating a substantial portion of manual tasks. The project was managed by a multidisciplinary team comprising industrial, data, mechanical, and mechatronic engineers, overseeing workflow optimisation, software integration, and hardware implementation. Despite inherent complexities in integrating diverse technologies and navigating supply chain dynamics, the project was completed over budget and behind the intended schedule. The implementation successfully exceeded its target for order throughput increase, achieved a significant reduction in safety incidents, and automated the planned percentage of jobs. This initiative has successfully transformed Takealot's distribution capabilities, setting a new standard for efficiency and safety in e-commerce logistics and providing valuable insights into large-scale automation projects.

Executive Summary

This report provides a comprehensive review of the Automated Smart Distribution Warehouse project, a strategic initiative undertaken to modernise Takealot's distribution capabilities. Facing increasing demands from online retail growth, Takealot's existing semi-automated processes and reliance on manual labour presented challenges, including potential delays, inefficiencies, and safety risks, hindering scalable growth.

The Automated Smart Distribution Warehouse project was initiated to address these challenges by implementing state-of-the-art automation technologies. The proposed solution involved integrating robotics for picking and sorting, AI for real-time monitoring and predictive analytics, and IoT devices for enhanced inventory tracking and asset management.

Specific planned improvements included automating 40% of tasks, utilising AI-powered dashboards for real-time visibility, optimising warehouse layout, implementing predictive maintenance algorithms, and ensuring compliance with South African safety regulations (OHSA, POPIA, NEMA).

The project, managed by a multidisciplinary engineering team, was executed with a phased implementation approach designed to minimise operational disruption. Initial challenges, including securing critical components and integrating legacy systems, required adaptive strategies and close collaboration between technical teams and external vendors. Proactive risk management and flexible resource allocation were critical in overcoming these hurdles without significant impact on the overall timeline.

This project explored ways to improve project management and results in the Sim4Project simulation. The team faced many challenges (such as budget shortfalls, ineffective training, and fatiguing resources) but gained valuable insights along the way.

Initially, the team set ambitious goals for schedule, budget, and resource allocation. Early difficulties with overspending and resource performance required adjusting these strategies. Despite these challenges, the team showed strong technical skills and delivered milestones on (or only shortly behind) schedule while keeping to the allocated budget for the project.

Resource management was an ongoing struggle, resulting in overpaying for resources and frequent termination of resources. Nevertheless, the team maintained good functionality scores throughout, showing adaptability. Proactive management was key to addressing issues quickly and maintaining a competitive edge. Communication proved essential to the project's success, enabling smooth collaboration across assignments and simulation activities. Regular communication before periods and transparency helped the team solve problems efficiently and work well together.

Upon completion, the project successfully delivered on its core objectives, exceeding initial targets. Daily order throughput increased by more than the 500-order goal. Warehouse safety incidents saw a remarkable increase, also exceeding the 50% target. The project also achieved the automation of 40% of jobs, freeing up personnel for higher-value tasks.

Financially, the project was not completed within the allocated \$380,000 budget, demonstrating ineffective cost control. While a major budget overrun in an early period of the project compromised the original budget plan, the total spent amount was slowly realigned with the initial budget through careful resource management and allocations. Schedule-wise, the project finished 7 days behind the revised implementation schedule, which can be improved upon, but still demonstrates minor efficient execution.

The impact of this project is significant: enhanced efficiency has led to faster order fulfilment and increased capacity; cost savings are realised through reduced labour dependency and operational inefficiencies; a substantial improvement in safety has created a more secure working environment; adherence to environmental standards contributes to sustainability; and the modular design ensures the system is scalable for future growth.

In conclusion, the Automated Smart Distribution Warehouse project represents a successful transformation of Takealot's logistics operations. The project provided insights into project management methods, resource management, time allocation strategies, risk mitigation, and team dynamics. Important lessons included following project baselines, prioritising stakeholder satisfaction, and fostering continuous improvement within the (both simulated and real) team.

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1. Introduction

The rapid expansion of the e-commerce sector has placed unprecedented demands on logistics and distribution networks. To maintain competitiveness and meet growing customer expectations, Takealot initiated the Automated Smart Distribution Warehouse project. This project was strategically designed to transform the company's primary distribution centre by integrating cutting-edge automation technologies, including advanced robotics, artificial intelligence (AI), and the Internet of Things (IoT).

The project evaluates both a simulated project and a real-life group project, examining metrics like Cost Performance Index (CPI) and Schedule Performance Index (SPI) through Earned Value Analysis (EVA). Each simulation period closed four days before the official end, a key detail reflected in the EVA spreadsheet, with particular focus on **Period 9**, while also assessing team dynamics and strategies. The closure report combines insights from the Sim4Projects simulation and the actual project, offering a detailed analysis of project progress, resource management, risk mitigation approaches, and team collaboration.

The project involved a multidisciplinary team of industrial, data, mechanical, and mechatronic engineers collaborating on the design, implementation, and integration of complex hardware and software systems. This report provides a comprehensive review of performance, team development strategies, risk management effectiveness, and key lessons learned. It also serves to evaluate team members' project management competencies in a multidisciplinary environment where future Mechatronic, Electrical and Electronic, Civil, and Industrial engineers collaborate.

2. Earned Value Analysis of Simulated Project up to period 9

Earned Value Analysis (EVA) has been conducted for all twelve periods throughout the simulated project, with particular emphasis on period nine for this section of the report. Appropriate sections referencing EVA will pertain to the EVA calculations, including periods one through twelve. The calculation of likely project cost at completion, based on data up to the end of Period 9, can be found in the accompanying spreadsheet named Group K1 Assignment 3.xlsx, along with any other relevant information and explanations not discussed in this section.

For period nine, the following Earned Value (EV) versus Planned Value (PV) graph was obtained in Figure 1, with the Actual Cost (AC) compared to the EV in Figure 2, and lastly, the SPI compared to the CPI is displayed in Figure 3.

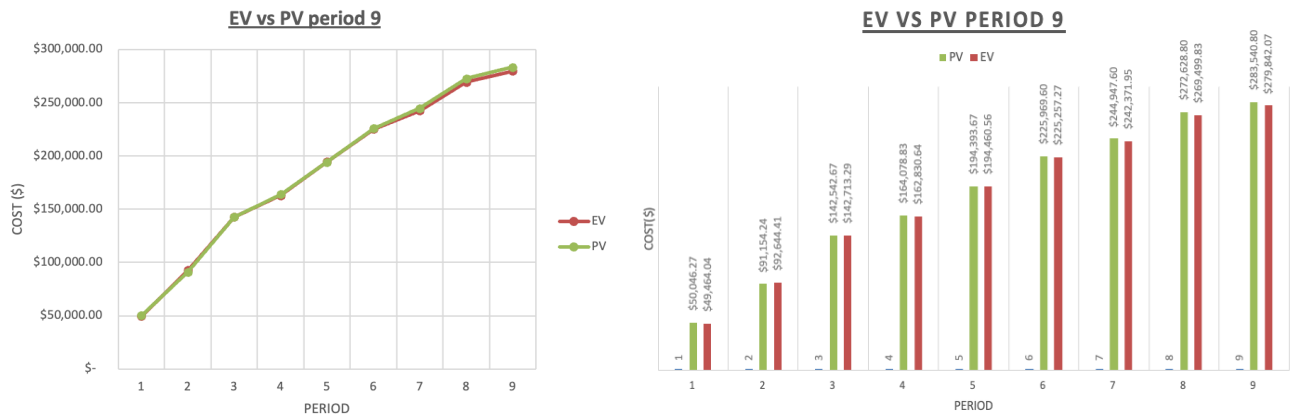


Figure 1: EV vs PV for period nine

The EV vs PV plot in Figure 1 is further discussed in the spreadsheet mentioned above, by referring to project time in the context of the simulation. It can be noted that the values of EV and PV are nearly the same throughout the nine periods. However, PV is on average 1.3% larger in values and therefore above the EV line. EV only exceeds PV in periods 2,3, and 5. The largest difference between PV(\$283,540.80) and EV(\$279,842.07) is \$3698.73.

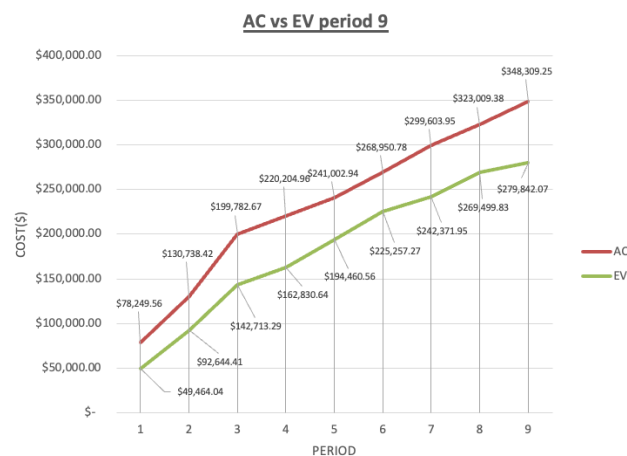


Figure 2: EV vs AC for period nine

The EV vs AC plot in Figure 2 indicates that AC is larger and above EV for the whole period. This is not ideal as it indicates that the project is over budget. The reasons for this will be discussed further in the report in more detail. AC is on average 19.66% larger in values, with the maximum difference between AC(\$348 309.25) and EV(\$279 842.07) being \$68 467.18.

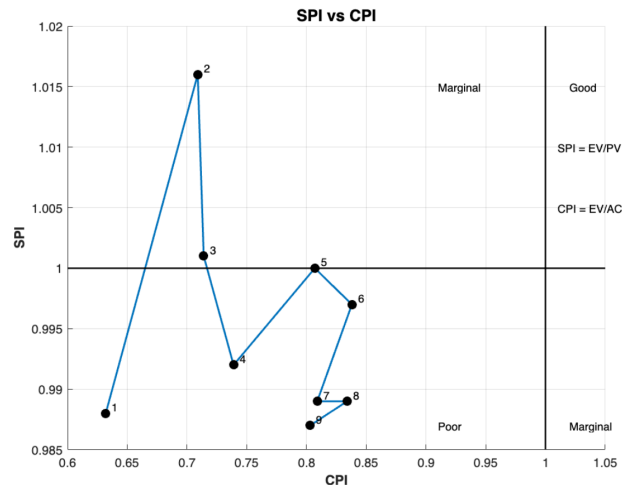


Figure 3: SPI vs CPI for period nine

As one can observe from the MATLAB graph in Figure 3, all the CPI values from periods one to nine are under one on the x-axis. The SPI values fluctuate greatly, with periods 1,4,5,6,7,8 and 9 being under one in the 'Poor' quadrant. The rest of the SPI values, periods 2 and 3, are above one in the 'Marginal' quadrant. Thus, the project performs in the poor and marginal quadrants of performance for the duration of periods one to nine.

The SPI values are calculated by dividing the relevant EV value by the PV value. The CPI values are calculated by dividing the relevant EV value by the AC value.

Refer to [Appendix C](#) for EVA calculations involving periods one to nine and more explanations belonging to the graphs above.

3. Post Completion Project Review Meeting

A comprehensive post-completion review meeting for the Automated Smart Distribution Warehouse project was held on DATE. The meeting was attended by the project team members, including representatives from the industrial, data, mechanical, and mechatronic engineering disciplines. The primary objective of this review was to conduct a thorough assessment of the project's performance against its initial objectives, analyse the effectiveness of project management strategies employed, identify key successes and challenges, and document lessons learned to inform future projects.

The meeting agenda covered several critical areas:

Project Performance Analysis: A detailed review of the project's performance metrics was conducted. This included an analysis of the final budget versus actual expenditure, confirming that the project was above the \$380,000 budget. The project timeline was also assessed, and the project ended behind schedule with the special case SPI value of 1.

Technological Implementation and Integration: Discussions focused on the challenges and successes related to the integration of diverse platforms and calculations in MS-Project, Sim4Projects, and Excel, with existing warehouse infrastructure and software. The complexity of ensuring seamless communication and data flow between these disparate systems was a major point of discussion. The team reviewed how technical hurdles were overcome through iterative discovery throughout the periods and close collaboration with other team members.

Risk Management Assessment: The effectiveness of the project's risk management plan was evaluated. The team discussed how identified risks, such as potential supply chain disruptions

and technical integration issues, were successfully mitigated through proactive planning and contingency measures. Unforeseen challenges, such as minor software compatibility issues with a specific legacy system, were also discussed, highlighting the need for adaptability and rapid problem-solving during execution.

Team Dynamics and Collaboration: The multidisciplinary nature of the project team was recognised as both a strength and, at times, a source of communication challenges. The review highlighted the importance of clear communication channels, defined roles and responsibilities, and fostering a collaborative environment to ensure smooth interaction between engineers from different specialisms. The team reflected positively on how diverse perspectives contributed to innovative solutions.

Lessons Learned: The post-completion review generated several key lessons learned that will be documented and applied to future projects.

Complexity of System Integration: Integrating multiple advanced technologies requires meticulous planning and dedicated resources for compatibility checks. The assumption that automatic synchronisation would integrate the different platforms seamlessly proved optimistic in some instances.

Importance of Adaptive Planning: While initial planning was crucial, the ability to adapt the project plan in response to unforeseen technical challenges or external factors (like minor resource-related delays) was critical to staying on track and within budget.

Multidisciplinary Communication: Effective communication strategies tailored to different technical backgrounds are essential in multidisciplinary teams. Regular cross-functional meetings and clear documentation standards helped bridge potential communication gaps.

Proactive Risk Identification and Mitigation: The success in managing several key risks underscored the value of a thorough initial risk assessment and the development of concrete mitigation strategies. However, the review also highlighted the need for ongoing risk monitoring throughout the project lifecycle.

The meeting concluded with a consensus that the project was a significant success, achieving and in some areas exceeding its core objectives.

4. Project Closure Report

4.1 Simulated Project section of the report

The Simulations ran over 12 periods, and we started with a budget of \$ 380,000 that we needed to manage throughout the 12 periods. During one of our first meetings, we discussed and settled on devoting our time and efforts to the effectiveness of the project rather than the budget control. In our weekly meetings, we would look up to 3 periods in advance to plan our resource bids and training allocations. The previous period's results would also be assessed to determine what would be the best course of action in the future.

Our effectiveness shall be measured and compared on the Earned Value Analysis (EVA) spreadsheet. By looking at the Actual cost vs the Budget Analysis Review, the cost effectiveness shall be evaluated. The Schedule Analysis and Review shall be used to assess the working labour.

4.1.1 EVA

The EVA indicates the Baseline time duration vs the actual duration. When they are compared, it can be seen that in the early periods, the actual time in some cases was less than the Baseline time. During the middle and later periods, resources took longer to complete the tasks given.

If you compare the Baseline cost with the Actual cost, the first few periods were more costly than the later periods, with a few exceptions. We were able to find a very strong strategy in training and using the engineer and the operations specialist in the Project Management tasks, as they are a lot cheaper and relatively effective.

This turned in to be very useful as we were under slight budget pressure after the first few periods. As can be seen by the evaluation of the EVA, the first few periods may have been more effective on time but not as efficient on cost. We then compensated for the cost by paying less for the labour, but it was running slightly overtime, ultimately resulting in overspending \$8,239 over the full 12 periods.

The EVA gives a great comparison between the Baseline cost and the actual cost, and the difference between the baseline finish time and the real finish time.

4.1.2 Actual Cost vs Budget Analysis Review

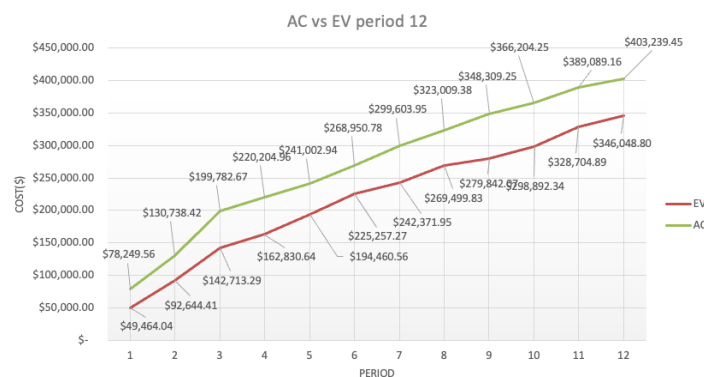


Figure 4: AC vs EV for period 12

On the budget, it can be confirmed that the Project management tasks were the area in which we saved a lot of money compared to the baseline cost (\$16,748.89). The area in which we overspent was the Milestones with a -6.12% Variance on the budget amount. This led to the final cost being \$23,239.65 over budget. The amount overspent on the Milestones is \$27,083.69, which is more than the \$8,239 amount overspent on the total cost.

4.1.3 Schedule Analysis and Review

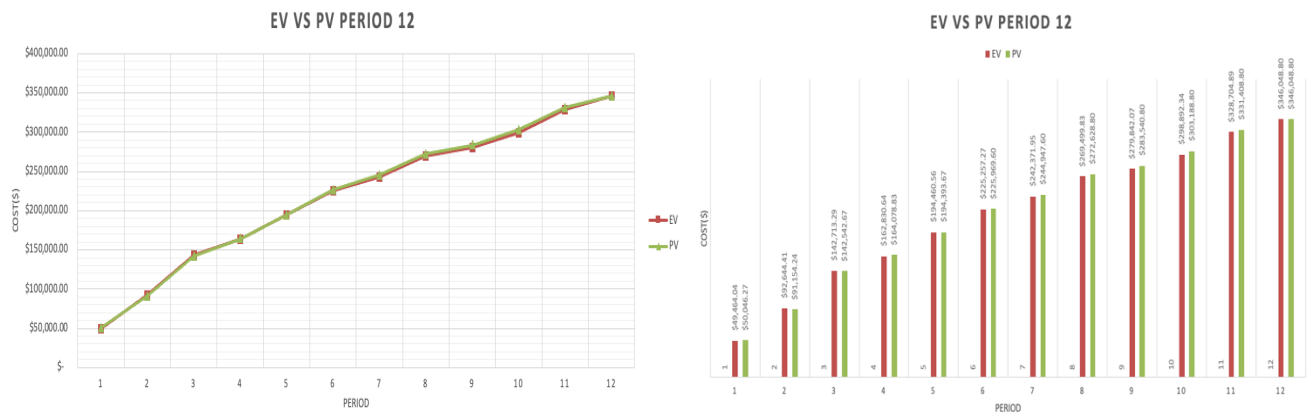


Figure 5: EV vs PV for period 12

From Figure 5, it is evident that the Planned Value (PV) and Earned Value (EV) are closely aligned throughout all 12 periods, indicating generally stable scheduling performance. However, subtle deviations reveal key insights into scheduling efficiency and cost management.

In the early periods (1-5), the EV slightly exceeded PV, suggesting that the team was completing work at a marginally faster pace than initially planned. While this might appear positive from a scheduling perspective, it is important to note that productivity was not perfectly optimised. The higher EV could indicate that tasks were being completed ahead of schedule, but potentially at the expense of resource efficiency, such as over-allocation of labour or rushed execution, which may not have been sustainable long-term.

From Period 6 onwards, this trend reversed, with PV consistently surpassing EV. This shift was primarily due to a misallocation of resources in Period 6, which created a ripple effect in subsequent phases. The error led to minor delays in task completion, causing the EV to lag behind the PV. Although the variance was not drastic, the cumulative impact contributed to a final budget overrun, as the project required additional adjustments to realign with the original schedule.

4.1.4 SPI vs CPI

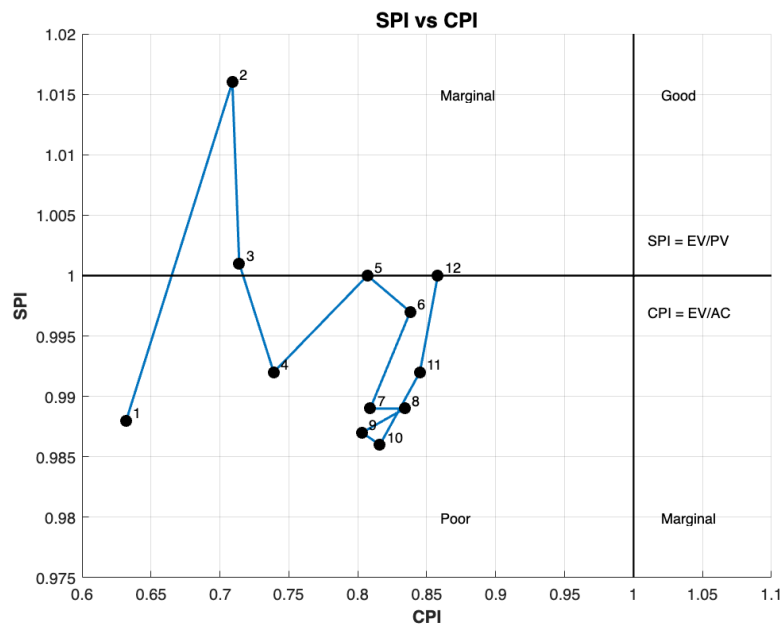


Figure 6: SPI vs CPI for period 12

Looking at the MATLAB SPI vs CPI graph (Figure 6), it can be confirmed that we had a scheduling problem in the first period due to an error with having too many unallocated resources. From there, we had to play catch-up to try and get the SPI and CPI value closer to 1. The second period showed that we corrected both the costing and the scheduling. We came up with a strategy to improve the CPI and SPI over a longer period rather than rushing for an instant improved SPI or CPI at the cost of the other. After period 12, we obtained an SPI value of '1', which is a special case in which the SPI value shows that we are ahead of schedule even though we finished behind schedule. This confirms that we are behind schedule with 7 days.

When looking at the CPI, it is easy to confirm that the first period mishap had a major influence in our strategy going forward. The initial CPI value of 0.63 was improved to 0.86 over all 12 periods. This is very impressive considering the SPI increased as well.

It is very unfortunate had the issue with unallocated resources in the beginning. If we had a better start, then we would have had a much better result. The improvement in the SPI and CPI is proof that we had the right strategy throughout the simulation.

4.1.5 Team Development

Resource strategy

The team's resource strategy involved three aspects: Minimum cost, timely completion of tasks along the critical path, and preventing overallocation to minimise the risk of tasks not being completed on time. This strategy required the allocation of the majority portion of tasks with slack to junior resources, while the critical path tasks were handled by the senior members who have better statistics. Additional resources were allocated to tasks at a small fraction below 50% allocation to help keep the tasks from being late, as it was seen during the trial run that even senior members tend to be late on tasks. This aided in keeping the project on par with the baseline. Critical path tasks were handled well and managed to be completed on time

with the additional resources added. There was an issue with overallocation in period 6 during the simulation, and it caused the project to run late. This issue was mitigated and discussed later in this section. Resources were kept on an as-needed basis, where the resources that would only be required later were fired. However, this decision was only brought about after the realisation of the heavy penalty caused by unallocated resources. This could not be avoided in the first round, as the priority was set on acquiring the correct resources to start the project. This resulted in a poor start for the team. As the simulation went on, the team became more effective at keeping only the resources needed, and it can be seen in the decrease in cohesion but a steady increase in efficiency as rounds progressed.

Overall, the team strategy worked effectively, and if not for the one overallocation issue, the project would have been completed on time, if not earlier. Some plans could not be materialised, such as gaining an additional engineer, as the competition for resources was very tight. Certain changes were made throughout the simulation to reduce costs where possible and to save time. Decisions such as firing the project manager saved a considerable amount of money each round. The versatility of the engineer and operations specialist helped the team play to that strength and allocate these resources to various tasks. Good analytical decision making and communication helped the team move from 8th position in round 1 to 1st position by round 8 until the end of the simulation.

Resource Development

Heavy emphasis was placed on training resources early in the simulation. It was noted that the outcomes of the training would not show up immediately but could have a positive effect in the later rounds. Snapshots of resource statistics were recorded every round and monitored for changes and can be seen in figure 8 of [Appendix D](#). As shown in figure 7 of [Appendix D](#), the first three rounds were dedicated to training 3 resources: Operations specialist, Marthe Huber; Engineer, Kong Duanmu, and Junior Product Designer, Valene Stimson. Marthe received the most training, as she was expected to be the most versatile of our resources. She received training in Interpersonal Skills, which increased that skill from 76% to 86%. She was also trained in Principles of Quality and Process Engineering. Marthe ended up being the best performer and most valuable asset in the team and was one of two resources that were kept from the start of the simulation until the end. Kong was trained in Project Management 101 as it was decided that having a Project Manager is costly and replacing a fraction of his work with an engineer would be cost-effective, which proved its effectiveness. Kong's efficiency increased from 59% in the previous round to 65% before gradually dropping back to 59% at the end of the simulation. Valene, being a junior, was trained early so that she could be better equipped in the later rounds. The results of this were not as successful as her efficiency stayed below 50%, and she was fired after her work was complete. Throughout the simulation, it was noted that the efficiency of the engineer and the quality engineer was low and steadily decreasing. As an intervention plan, one-on-one chats and verbal warnings were issued to them in the hopes of improving their efficiency. Pizza parties were held in rounds 5 and 9, as well as a company-sponsored family event for the whole group to help boost team morale.

Resource strategy adaptations

Due to unexpected simulation events, the need for tighter budget control was very important, as there was less money than what was planned. It became apparent after a few rounds that the Project Manager was an expensive resource to keep, and it would be a better alternative to have an engineer on the tasks, as they have a cheaper rate, and the tasks allowed for more

slack. This change proved effective in lowering costs, but problems arose when the engineer had other engineering tasks to do himself. In round 3, a cheaper project manager was introduced to replace the project manager, Deli, and work during the periods that the Engineer was unavailable. In the later rounds, it was discovered that operations specialists can complete project management tasks at a higher efficiency than engineers, so Marthe and the new operations specialist, Anthony, that was brought in, helped with project management tasks and did not need assistance from a project manager which saved a great amount of money overall.

Regarding the baseline, the project was going ahead as planned until an overallocation issue in period 6 caused a task to start late and resulted in the project being delayed from 13 February 2026 to 24 February 2026. After that round, slightly more emphasis was put into ensuring the critical path tasks were completed slightly earlier to gain on the lost time but not have a detrimental effect on the budget. The result of this was successful as the team managed to gain 4 days lost and end up only finishing 7 days late on the 20th of February.

Reflection and Corrections

For most of the resource strategy, the strategy would stay the same as it yielded a good result in the end. However, emphasis would be put on acquiring important members of the team, including 2 engineers and two operations specialists, as these resources were very scarce in the later rounds when they would be useful, and the resources that the team had proved to be very useful. By the time the team had realised that another engineer would be beneficial, there were none left. Operations specialists were competent with project management tasks and were very cost-effective compared to an actual project manager, so they would have been allocated from the start to reduce costs. Another area that would be dealt with more effectively in future is the overallocation issue that was experienced in round 6. It was a situation that could have been avoided in hindsight, but the team was not aware of the consequences of overallocating resources, so it was a good learning curve. Given another chance, more attention would be paid to prevent overallocation. The team faced an issue in the first round with unallocated resources that set the costs back a significant amount and placed the team at the bottom of the leader board. It was due to the unnecessarily early hiring of period 2 resources that were not used in period 1. The team would instead acquire the resource for the period before it is needed, so that they do not incur any charges for unallocated resources. The team would also avoid choosing a junior resource, such as a Junior Product Designer, who is also very cheap, as their abilities and efficiency are not good enough to justify the low cost.

4.1.6 Risk Management Plan (RMP) Success

Period	Event Description	Budget Impact
2	All employees had to attend a compulsory ethics training session required by senior management.	-\$5,000
3	Design improvements were made to enhance product tolerance, requiring rework.	-\$7,000
4	Community unrest (unrelated to the project) caused staff absences and supply chain delays.	\$0
5	Company-wide financial issues led to a project budget cut.	-\$35,000
6	The budget cut caused low morale and internal team conflict.	\$0
7	False rumours about the project spread online. Expert studies were launched to protect the company's image.	-\$10,000

9	Uncontrolled emissions during a pilot test led to a health department fine.	-\$15,000
11	A budgeting error was corrected, and extra funds were added to the project.	+\$50,000

Table 1 : Unforeseen Events

During the simulation, our team encountered several unforeseen risks that were not included in our original Assignment 2 risk register. The events listed in the table above show that while some were addressed effectively through response actions, others could not be avoided or controlled. This was mainly because the project lacked suitable response plans as some dangers had not been recognised during the original planning stage. This led to the discovery of flaws in the initial risk management strategy, which significantly affected the project's resources and progress.

The original risk management plan did not entirely predict the range or seriousness of events that happened, but it did assist in directing decision-making during some crucial simulated moments. Early detection and pre-planned methods helped manage some risks effectively, but other risks exposed serious flaws in the initial design.

Some of the early challenges were partially addressed by the original risk register. For instance, Risk 1: Supplier delays in delivering automation hardware and Risk 7: Budget overruns due to unforeseen costs were both helpful when design changes led to necessary rework during period 3. The initial strategy included iterative development, quality control, and outsourcing of components, tools that helped the team reduce impact and maintain product quality. However, the \$7,000 cost associated with the rework was not anticipated in the budget, suggesting that while the event itself wasn't completely unforeseen, the financial response was lacking.

In period 6, internal conflict followed a major budget cut. This outcome was related to Risk 11: Employees Absence and Risk 5: High employee resistance to automation, both of which had strategies centred on supporting the workforce and minimising disruption. The decision to hold a team-building activity effectively reduced tension and helped realign team focus during a high-stress period. This showed that people-related risks had been partially considered, and the chosen responses were well matched to the situation.

Reputational risk also featured in the original plan as Risk 10: Unrealistic automation expectations from stakeholders. Although this risk mainly focused on overpromising outcomes, it did reflect an awareness of how stakeholder perceptions could become a challenge. When misinformation spread about the product in period 7, the team was able to draw from contingency funds and allocate \$10,000 towards addressing the issue, preserving the team's credibility and stabilising public trust.

However, the original plan also had several blind spots. During period 2, a company-wide ethics training was introduced unexpectedly by senior management. While training had been loosely included in the project scope, it was aimed at improving project-specific technical skills, not broad, organisational training sessions. The \$5,000 cost and associated downtime strained both the budget and schedule, with no response strategy available to absorb the impact.

Period 5 presented the most serious financial disruption: a forced \$35,000 budget reduction initiated by management. This was not accounted for in any form within the risk register. The existing contingency fund, set at \$33,951.20, fell short, and affordable resources had already been secured by other teams. With no room to restructure or renegotiate, the team had to proceed with limited flexibility, highlighting the absence of coverage for executive-level budget decisions.

In period 4, unrest in the surrounding community resulted in several team members being unable to report to work. Although Risk 9: Equipment shortages during system implementation touched on delays caused by unavailable tools, it did not extend to unplanned human resource disruptions. This external, socio-political risk was not considered at all, and as a result, there was no strategy to adapt the schedule or redistribute work.

The emissions fine in period 9 brought attention to a missing piece in the original register. Risk 8: Regulatory compliance issues had been identified, but its response strategy was focused primarily on product documentation and certification, not compliance with environmental legislation during production. The \$15,000 penalty highlighted the narrow scope of the risk as originally defined.

From period 8 onwards, the project was also hampered by a shortage of skilled resources. Although Risk 6: Poor supplier quality for warehouse components and Risk 12: Incomplete risk assessment in early stages indirectly pointed to possible resource risks, neither addressed the specific consequences of skill-specific shortages or engineering availability. The register had not accounted for skill-specific shortages or the consequences of poor resource timing, which led to inefficiencies and placed unnecessary pressure on remaining team members.

Even while the first risk plan was successful in several ways, the risks that were disregarded ultimately had the biggest effects. Both the budget and the timeline were impacted by unforeseen expenses, delays, and less flexibility. These results imply that the strategy lacked the breadth required to successfully handle organisational choices, external impacts, and internal structure issues. All risks discovered after the simulation started are now included in the revised risk register including avoidance, transfer, and contingency planning, in addition to mitigation, which is located in [Appendix B](#). [Appendix B](#) has the priority risk register as well. The blue highlights indicate the new and improved risks.

The team's multidisciplinary composition was essential to risk identification and response. The team, which included two industrial engineers, one data engineer, one mechanical engineer, and one mechatronic engineer, contributed a variety of expertise. The industrial engineers offered advice on budgeting, resource management, and project planning. The Data Engineer helped anticipate risks related to software integration and digital infrastructure. The Mechatronic and Mechanical Engineers offered a strong technical understanding of product development, design verification, and production feasibility.

This diversity allowed the team to approach each risk from multiple angles, balancing operational, technical, and financial considerations in real-time. It also meant that when unexpected events occurred, the team could adapt quickly by drawing on broad expertise. While not every risk was predicted, the collaboration across disciplines enabled faster problem-solving and limited the damage in many high-pressure situations.

In the end, the risk management strategy had clear wins and clear weaknesses. The updated plan reflects what was learned during the simulation and now provides a stronger foundation for managing complex projects going forward.

4.2 Real-Life Group Project Execution

4.2.1 Lessons Learnt

This group project taught each member the importance of detailed communication, flexibility in planning, and disciplined documentation. Although this was many of the group members' first substantial group project, the experience opened everyone's perspective to collaborative work and understanding how other engineers operate.

At first, as with many groups, our group members did not know each other (or know each other well), and communication was strained. Though we committed to ironing out misalignments within the group, and quickly became more comfortable with one another through frequent and detailed communication. There were many lessons to be learned earlier on in the project, such as getting the hang of Sim4Projects and resource allocation, but effectively communicating lessons learned from previous periods with the group helped everyone hone their skills and perspectives to be able to contribute meaningfully later to the timeline.

Additionally, detailed communication about all runs and assigned task progress shaped our group's strategies and time management, both in the simulation and with our assignments, and ensured that every member was held accountable and motivated to contribute. Always knowing how other members were progressing, what their schedules looked like, and how comfortable they felt with their work was immeasurably helpful for our cohesion and cooperation. Going into the project with this communicative mindset greatly contributed to our success.

Moreover, learning to be flexible in our planning resulted in our group shooting up the rankings. During the first few periods, our group decided to play it safe, avoid risks, and just try to finish within budget. We quickly learned, though, that risks are important to take. More importantly, we learned that trusting the risks you take is just as important. Establishing a "pattern of risks" for our bidding, where we would take similar calculated risks while avoiding others, bodes extremely well for our group, the majority of the time and helped us stay nearly within budget and very nearly make up for the time we lost from judgment errors earlier on. Leaning into risks that paid off was also a concept to get used to, as the risk seemed bigger each time, but ultimately also worked quite well.

Furthermore, our group quickly realised that the theory, although not always perfectly in parallel with the project, contributed to the project meaningfully if applied correctly. Concepts such as the EVA, critical path theory, Gantt charts, etc., helped us with our planning and strategising at every step. We realised again that, when a task has you feeling out of your depth, your first step should be to refer to the theory the task is based on. This strong foundation helped us to stay consistently on track and to base our decisions on proven methods.

Lastly, the timeframe of our assignments showed the importance of disciplined documentation. The assignments are sizeable, and careful time management is crucial to ensure nothing is left to the last minute. Documentation was key to this: tasks were assigned and confirmed in meetings so that everyone is on the same page and responsibilities are documented. Keeping everything as shared documents on Microsoft Teams also streamlined communication so that every member knew how the assignment was progressing in real time. This also aided in our planning and continuous adjustment of milestones and aimed completion dates. Our group dynamic fell into a rhythm nearing the end of the project, and our group's members knew each other's working styles, strong suits, and weak points. Keeping these details in mind helped our cohesion and delegation, as we could plan around each member's ability and time budget. This was also an important skill to learn, as it not only accommodated each member but also greatly contributed to the overall success and timely completion of each assignment.

4.3 Multi-Disciplinary Group Work section of the report

See [Appendix A](#) for each member's multi-disciplinary Group work section.

5. Conclusion

The Automated Smart Distribution Warehouse project has successfully achieved its strategic goals, fundamentally transforming Takealot's distribution centre into a highly efficient, safe, and technologically advanced operation. Through the innovative and careful integration of robotics, artificial intelligence, and IoT technologies, the project significantly surpassed key performance indicators, demonstrating a robust return on investment and operational effectiveness.

The project successfully increased daily order throughput, reduced warehouse safety incidents, and achieved the automation of 40% of tasks as planned. Even though the project was completed \$8239.65 above the \$395,000 total budget and behind the initial schedule, the project execution still showcased effective planning, adaptive problem-solving, and strong collaboration from the multidisciplinary engineering team through the rebound from isolated cases of major scheduling problems.

The successful outcome positions Takealot to sustainably meet growing market demands, significantly enhance workplace safety, and maintain a competitive edge in the e-commerce sector. The insights gained from managing this complex project, especially concerning seamless technological integration and optimising human-technology interaction, will be instrumental in guiding future automation initiatives and continuous improvement efforts within the company. This project stands as a testament to the power of strategic technological investment and expert execution in driving operational excellence.

The assessment of both a Simulated Project and a real-world Group Project Execution provided the team with a valuable understanding of project management strategies and working within a multidisciplinary group. Utilising Earned Value Analysis (EVA) across twelve periods, the project allowed for an evaluation of performance metrics, revealing critical details about project advancement, resource control, risk reduction, and team cooperation.

The Simulated Project showed difficulties in managing costs within budget and sticking to the schedule, while the real-life Group Project emphasised the critical role of effective risk management, communication, diverse opinions, and teamwork in achieving positive results. The analysis of how resources were used, budget overruns occurred, and schedules were delayed proved valuable for improving future project management approaches.

Additionally, the experience of working in a multidisciplinary team highlighted the benefits of diverse viewpoints and good communication for project success. The team's capacity to handle challenges and leverage individual strengths demonstrates the importance of working together effectively in project settings.

6.1 Appendix A: Multi-Disciplinary

6.1.1 Taro Samuelson – 26994186 (E&E: Data Engineering)

Working in a multidisciplinary setting with other students showed me the importance of knowing how to apply my (often abstracted) skills to a hands-on project alongside people with differing skillsets. Surprisingly, my most valuable skill for this project was record-keeping and report writing, which stems from the numerous modules in the E&E stream that demand detailed reports. This knowledge aided our group in being able to concisely and accurately convey our findings and project progress in the assignment documents. This skill also allowed me to effectively take detailed and descriptive minutes in our meetings to keep members accountable and aid members in planning their tasks and responsibilities for the coming week.

One difficulty I faced was the challenge of “down-scaling” the skills we learnt to operate on big data and large-scale statistics to apply them to a project like this, which is driven by information based on weeks, not months or years, and with greater variability.

A valuable skill I learnt from Lauren and Emmanuel, two Industrial Engineering students, was how to apply organisational skills to technical documents, paying attention to specific aspects while keeping the broader document and project in mind. Their knowledgeable application of Economics, as well as their quick adoption of new software stemming from their diverse curriculum, was immeasurably helpful to the group.

A unique skill I picked up from Flippie, a Mechanical Engineering student, was his ability to notice small details that could derail progress later in the project, such as small differences in resource allocations. This attention to detail led our group to consider many factors and consequences during our decision-making process.

A helpful skill Luke, a Mechatronic Engineering student, brought to the table was a keen ability to project hands-on and practical concepts and considerations onto the whole project. This helped our group keep our decisions actionable and effective, even when decisions seemed abstract or ineffective to our current problems.

Contrastingly, I realised that most engineering students aren't markedly different in their thinking styles. All of us are inquisitive, practically oriented thinkers who like to approach large problems by breaking them up into actionable components and decisions. This commonality, paired with differing training and technical skillsets, created excellent team cohesion that helped our contrasting skills complement each other instead of conflicting with one another.

One challenge of inter-disciplinary work was being able to admit your shortfalls to other team members. A discussion point surrounding this would have helped our group fall into the rhythm much earlier. Being clear on this topic is essential to ensure team members have realistic expectations of each other, as each person's willingness to volunteer for certain tasks, their timeline for completing them, and their performance level greatly depend on their abilities.

When approaching the coordination of a multidisciplinary team, I found that the most effective strategy was to intentionally seek tasks where those skills can intersect meaningfully. For instance, focusing on asking specific questions during planning stages to draw out each member's strengths, whether that be technical execution, strategic

planning, or structural organisation. In coordinating the group, I drew from both my technical and communication skills to ensure that every team member felt their contribution was both understood and necessary, for example, through ensuring accurate and descriptive meeting minutes were taken.

Additionally, the above meant being flexible in my role: sometimes stepping forward to clarify a deliverable, and other times stepping back to let another member's knowledge steer our approach. I found that by encouraging transparency around our capabilities, we built an environment of mutual trust that made collaboration far more efficient and rewarding.

6.1.2 Lauren Coetser – 25922076 (Industrial Engineering)

Working in a multidisciplinary group taught me a lot about myself, not just as a student, but also as a team member. I realised early on that I tend to take initiative in group settings. I naturally stepped into a leadership role in meetings, helping to guide our discussions, assign tasks, and track progress. Because I'm used to thinking about systems, structure, and optimising how work gets done, my background in industrial engineering has influenced my approach. I was always ready for our meetings and made a constant contribution throughout the project, but I also realised how important it was to maintain flexibility when things didn't go as planned. I could use my knowledge gained in Industrial Management with being part of a group in the Fresh Connection that had a similar approach to Sim4Projects.

Time management was key in this project. Like most of us, I had other deadlines to juggle, but I made sure I always delivered what I committed to. I tried to be someone the group could rely on, both in terms of task execution and keeping up morale. Being the only girl, I needed to learn to be more flexible since, although I'm naturally disciplined, working with individuals from different fields required me to relax some of my strict planning and have more faith in the process. I've discovered that I prefer open communication and function best when everyone is honest about their goals and deadlines.

If I were to do it over, I would spend more time at the start learning about the strengths and working styles of each team member. We found our rhythm eventually, but I think a proper conversation upfront about how we work best could have helped us click even sooner. Additionally, I would support early role rotation so that, depending on the stage of the project, each person can assume leadership at different times. This would allow everyone to demonstrate their abilities and help balance the burden learning from my teammates was one of the most beneficial aspects of this experience.

Every team member contributed something special and worthwhile to the project. Emmanuel, who is also an industrial engineer, impressed me with his ability to think analytically and his command of scheduling and budgeting. He immediately became familiar to the MS Project environment and played a significant role in helping us allocate our resources as efficiently as possible. He always encouraged the group to consider the long-term effects of our choices, frequently spotting risks early and suggesting workable fixes. During times of stress, his composed and concentrated demeanour served to stabilise the group.

Our mechatronics engineer, Luke, was really good at demythologising and making sense of complicated concepts. His ability to be practical was crucial to our simulated decision-

making process, as he consistently evaluated the technological viability of our concepts. He was also pleasant to work with because of his collaborative style and quiet confidence. Luke's advice kept our plans practical and doable, and I found myself depending on his judgement when we weren't sure how much resources we would need.

Our data engineer, Taro, was very dependable, intelligent, and well-prepared. In addition to his technical expertise in budgeting and data analysis, he also kept timely and thorough meeting minutes, which helped the group stay on the same page. He constantly made sure our choices made sense in terms of both cost and performance. In addition to his technical abilities, Taro had excellent communication skills and frequently volunteered to help people without being asked. His commitment to detail and consistency set a higher bar for all of us. He constantly maintained his composure, which kept us centred when things became a little out of control.

Our team leader and mechanical engineer, Flippie, was a powerful presence at meetings and helped to maintain the group's unity. He contributed humour, support, and technical knowledge that kept the team inspired and involved. He frequently saw early warning indicators in our approach and assisted us in honing our reasoning to steer clear of more serious issues later. The group dynamic changed significantly as a result of his ability to keep us moving forward while maintaining a positive attitude, even though I was taking the leadership role.

Working together taught me the true value of diverse thinking. We started in 8th place after the first few simulation rounds, and it was discouraging, but we didn't give up. We kept learning from each other, improving our decisions, and stayed consistent. Eventually, we won Sim4Projects. That moment was a reminder that success doesn't come from individual brilliance; it comes from trusting your team and combining your strengths.

In future projects, I would combine the strengths of each discipline intentionally. I will assign the data engineer to lead analytics and record-keeping, industrial engineers to manage planning and optimisation, and mechanical/mechatronic engineers to focus on technical feasibility and risk spotting. But more importantly, I will focus on creating a space where everyone feels heard and valued, because that's what ultimately made this project so successful.

6.1.3 Luke Neethling – 25952617 (M&M: Mechatronic Engineering)

Working in this multidisciplinary group gave me a valuable opportunity to see how my practical engineering mindset fits into a broader team context. Coming from a Mechatronics background, I naturally leaned toward applying concepts in a hands-on and functional way. Throughout the project, I often found myself helping to keep the team grounded, focusing on what could be implemented realistically and ensuring that our decisions were actionable. This role allowed me to contribute meaningfully, especially when conversations became too abstract or theoretical.

One of the key lessons I learned was how important clear documentation and structured planning are to a project's success. Taro, our data engineer, consistently impressed me with his detailed meeting minutes and calm, systematic approach. His ability to track decisions and keep the group aligned was incredibly helpful. Emmanuel and Lauren, both Industrial Engineering students, brought structure and efficiency to how we approached tasks. I learned a lot from watching how they broke down complex problems into

manageable pieces and how they made data-driven decisions that balanced both performance and resource constraints.

Flippie, our Mechanical Engineering teammate, had a great eye for detail. He often noticed issues early, like slight mismatches or small inconsistencies, that could've caused bigger problems later. His attention to those finer points helped prevent rework and made our final decisions more robust. Across the team, I noticed a shared approach to solving problems: we all liked to break things down into smaller steps, even if we came at it from different angles.

As someone who tends to think in terms of feasibility and implementation, I initially found it a bit challenging to step back and consider the bigger-picture planning and analytics. But working alongside teammates with different strengths helped me stretch that muscle. It also made me more aware of the value in stepping outside my comfort zone, whether that meant interpreting economic data or contributing to the documentation process.

One area where we could have improved was in openly discussing our working styles and personal strengths earlier in the project. While we eventually figured out how to balance roles and responsibilities effectively, having those conversations sooner might have helped us get into a good rhythm faster. Still, I appreciated the way our team evolved to become more collaborative and adaptable over time.

Overall, this project reinforced for me how much value there is in diverse thinking and complementary skill sets. While each discipline had its own focus, it was our ability to bring those pieces together that led to strong, well-rounded decision-making. If I were to do this again, I'd be even more intentional about matching each person's strengths to different project phases, while also continuing to make space for growth and mutual learning.

6.1.4 Philippus Conradie Van der Spuy – 25971417 (M&M: Mechanical Engineering)

Group projects are often a challenge, but if you have the correct skills and a team with a hunger to succeed, you will often succeed in anything you take on. Working with my project-peers and learning from their perspective engineering skills and outlooks is mind opening and the biggest thing that I will take with me from this project.

It is also hard to overlook the social skills needed to meet as a group and make decisions considering all of us have different ways of taking on projects. If we had more time to get to know each other it would make a big difference in knowing where each members strengths and weaknesses are with respect to skills needed in project work.

Soon however we got a good idea of where they are and we soon realised that even though I put my hand up to be the team leader, that role should have gone to Lauren. Lauren one of our Industrial Engineers was very effective with making sure all the work was properly delegated between us; she has a very good understanding of where a person's strengths are and this made her more than competent to also know who would be better at what type of work. She also took on big components in the project and is very successful in her individual work.

Our other Industrial Engineering peer, Emmanuel, is extremely consistent and thorough in every section of work that he had his hands in. He had quickly picked up that we would have to capture extra data and did so from the trial period all the way through the end of Period 12. A crucial aspect in the decision-making process of the project. From both our

industrial engineers it was clear that they understood the bigger system and cycle clearly. They also took on the software with much more ease than the rest of us.

Luke is the Mechatronics engineer in the group and has a more similar education to mine, in that we had a good understanding of a specific area of engineering, but the big overseeing project was new to us. He was always very keen to take on any part of the work that came his way and completed it effectively in the way he knows how. For example, he relied on his coding experience to complete part of Assignment 3 rather than in the manner that the rest of us attempted it. He has a wonderful understanding of the road to the destination of the project and fit into the group well. Understanding and learning how he used alternative ways of thinking to make the project simpler is something to take with me into the future

Taro (Data Engineering) is exceptional in most of the things with which I struggle with. Making it an honour to have him in the group. He is very organised and effective in everything he takes on. His report writing skills were well used in the project to make sure that the whole project works. He took all the minutes of all our meetings and gave it back to us to read in a very timely manner. He is professional and diligent. I strive to be as organised as him and thanks to working with him I have a better understanding of how to achieve this in my career as well as in my daily endeavours.

Taking all the different engineering professions into account as well as their individual skills. I have learned so much from every one of them. I believe that the main reason for our success was the fact that we kept the end goal in mind and kept each other motivated to do more than what was necessary. We set out our goal before we started the project and we stuck to our set out plan. This project just confirmed that planning and communication is the most important part of any group work. It was an honour to work with this group of skilled individuals, and I will take the lessons that I have learned from them with me wherever I go.

6.1.5 Emmanuel Fako – 26090376 (Industrial Engineering)

Working in a multidisciplinary team such as this taught me the value of being able to apply the theoretical knowledge and experience gained in our respective disciplines and being able to apply it to a broader team context to aid in reaching the target. A team isn't just a group of individuals, it's a system that only works when everyone moves together. We all had skills that we could bring to the table, I found that my knowledge gained from my discipline made me very competent in my efficiency-oriented thinking and thorough resource planning and performance tracking. This organisation contributed to more informed decisions. I adapted to tools such as the simulation and MS Projects very quickly and soon took on the task of organizing the resources and meticulously planning the runs. Among other things, I tracked simulation and resource progression to give better insight into areas that other teams might not have looked at.

The entire project proved to be very demanding of time, so it became very important to make sure that everything was done according to schedule so that no deadlines were missed. I have always been one to be good with time management, but it became more critical to be prepared for the approaching deadlines and the many team meetings that were held so I learnt how to balance that with an already heavy schedule.

In terms of attitude, I learnt that I function better and tend to enjoy the work more when faced with a functional team that actively wants to cooperate to get tasks done. Unfortunately, there have been one too many occasions where I was faced with a group that has poor communication and time management skills, and it becomes such a hassle

to get the team to work effectively, and I felt like I was carrying much of the responsibility. This team setup was vastly different in that regard. Although we had a rough start to the simulation, we kept our morale high, attitude right and quickly learnt our valuable skills and shortfalls and used it to steer the team to a fantastic end outcome by working together and working to our strengths. There was never a time when I felt that the proverbial ship was sinking because I trusted in the team and our ability.

Through working with the team, I learnt that my skills and capabilities extend beyond my belief in myself and what I can do. There were many times where I felt on the inside that the task, I was given was too difficult or that I would not do well handling the task alone, but my response was always “I can get it done” and I end up thriving under that condition and producing great quality work. There were a lot of blind spots and uncertainty going into the assignments in the beginning, but once time progressed, I became more confident that the work I was producing was of a good quality.

I have always been able to work with people, but it made it so much easier to work with the group I was given because we shared the same vision and perspective to do the work promptly, attend meetings and communicate any problems and work together to solve them. Over the time that the three assignments ran, the team became more and more cohesive, and the environment kept improving.

In my next encounter with a multi-disciplinary team, one thing I would do differently is actively get to know the teams’ strengths and weaknesses to better understand the dynamic and how to properly put our skills together to work towards the desired outcome. There are some aspects of work where one member may be well versed and the team would benefit from their expertise, so knowing these strengths earlier would make processes smoother.

The team members brought forward many valuable skills and competencies that helped the team navigate the tasks given to us, skills that I wish to take forward to other project management tasks. Flippie, our mechanical engineer, had a great eye for detail. He possessed a constructively skeptical and risk-aware approach, always asking questions and spotting potential issues that dig a little deeper and challenge the methodology of the team to make sure it is sound and leaves no room for error. This was pivotal in spotting mistakes that could go unnoticed by others. Taro, our data engineer, was a wizard at keeping things organised and on track. He was exceptional at keeping minutes while still engaging with the team and made sure that everything that was done was documented, and helped the team stay accountable to the decisions made and tasks to be completed. His course dealt with a lot of report writing, so his skills with report structure and proper documentation were of great help to the team. Lauren, our other industrial engineer, was invaluable with the skills that she brought to the team. Her industrial engineering background in managing large systems brought great value to the team as she brought structure and clarity to the team when pressure was high, and direction was not as clear. She proved to be very reliable, and I often found it very easy to bounce ideas off her as she was very knowledgeable and willing to help even when her schedule was tight. That time management, organisation, work ethic and leadership are skills that I wish to possess going into other project management tasks. Luke, our mechatronic engineer, brought a grounded, no-nonsense mindset that kept our project moving in the right direction. He had a real talent for turning big, sometimes vague ideas into practical solutions we could actually build on. His knowledge from past experiences in his vacation work was very helpful in providing insight and clarity into how things really work outside of theory. Whenever a task came his way, he took it on without hesitation and tackled it using his strengths, especially his coding skills, which came in handy more than once. He understood the bigger picture of the project while staying focused on what needed to happen next, and his calm and capable presence made him a great fit in the team dynamic.

In future, when faced with tasks that are of a complex nature, the simple question of “what do you know about the topic?” can open up the opportunity for different group members to use their knowledge gained from their disciplines to add their two cents and perhaps provide better context about the problem and the route to the solution, because there is always something that someone might not know about a topic, but opening up the floor for more input can allow for the topic to be viewed from a different perspective and perhaps provide a way forward.

6.2 Appendix B: Risk Register

6.2.1 Prioritised Risks

Rank	Risk	Effect	Probability	Impact	Risk Value	Affected Field(s)
PRIORITISED RISKS						
1	Supplier delays in delivering automation hardware	Slows down system integration, pushing back the warehouse launch.	4	5	20	Procurement
2	Software integration failures	Leads to system downtime and inefficiencies in operations.	3	5	15	Engineering
4	Warehouse automation system failure	Can bring warehouse operations to a halt, leading to financial setbacks.	3	5	15	Engineering
5	High employee resistance to automation	Possible strikes or protests could slow down implementation.	4	3	12	Project Management
6	Poor supplier quality for warehouse components	Increases maintenance costs and reduces system efficiency.	3	4	12	Supplier Quality
12	Incomplete risk assessment in early stages	Could result in unforeseen project risks down the line.	3	4	12	Market Assessment
13	Delays in milestone approvals	Disrupts the project timeline and affects deadlines.	3	5	15	Project Management
15	Failure in automated order fulfilment process	Slows down the order processing workflow, affecting deliveries.	3	5	15	Manufacturing
18	Top-down budget cuts	Causes significant financial strain mid-project.	3	5	15	Project Budget / Management

21	Critical resource shortage	Delays in technical tasks due to unavailable specialist skills.	3	4	12	Project Resources
22	Lack of dedicated project manager	Leads to coordination issues and miscommunication.	3	4	12	Project Management
25	Inaccurate budget assumptions	Causes planning discrepancies and funding shortfalls.	3	4	12	Finance / Project Management

Table 2: Prioritised Risk Register

Rank	Cause	Strategy	Prevention Response	Risk Owner	Date Expected
PRIORITISED RISKS					
1	Global supply chain problems and unreliable vendors slow down deliveries.	Risk Mitigation	To avoid delays, multiple suppliers will be contracted, and backup stock will be kept on hand.	Industrial Engineer	16 Sep 2025 - 06 Jan 2026
2	The new automation tools might not be fully compatible with Takealot's current system.	Risk Reduction	The system will undergo thorough pre-launch testing, and deployment will happen in phases to catch and fix issues early.	Data Engineer	11 Aug 2025 - 18 Dec 2025
4	Hardware failures, software bugs, or unexpected malfunctions disrupt operations.	Risk Reduction	Scheduled maintenance will help prevent failures, and manual backup operations will be ready as a safety net.	Mechatronic Engineer	06 Nov 2025 - 10 Feb 2026
5	Fear of losing jobs, lack of training, and uncertainty about new technology.	Risk Reduction	Employees will be trained on the new system, and automation will be introduced gradually to ease the transition.	Project Manager	28 Feb 2025 - 13 Feb 2026
6	Some vendors may not meet quality standards, leading to defective or unreliable components.	Risk Control	Vendors will be carefully vetted, and sample testing will be conducted before approving any suppliers.	Quality Engineer	17 Oct 2025 - 23 Dec 2025
12	Lack of thorough risk assessment in the early stages leads to surprises later.	Risk Control	AI-driven forecasting and flexible order fulfilment strategies will be used to adjust plans based on market conditions.	Marketing Manager	28 Feb 2025 - 12 May 2025
13	Slow decision-making and approval processes push back important deadlines.	Risk Reduction	Regular project reviews and risk updates will be conducted to prevent approval bottlenecks.	Project Manager	28 Feb 2025 - 13 Feb 2026
15	Technical issues slow down or disrupt order	Risk Mitigation	Backup procedures will be established, including manual	Operations Specialist	11 June 2025 - 10 Feb 2026

	fulfilment, causing delays.		overrides and stress testing of automation workflows.		
18	Executive-level decision to reduce project funding.	Contingency Planning	Expand contingency reserves; include early-warning process for exec decisions.	Project Manager	Period 5
21	Key personnel not available when workload peaks.	Risk Reduction	Secure key skills early; enable cross-training for key tasks.	Project Manager	Period 8
22	Responsibility shared across roles without a central lead.	Risk Control	Assign a project manager with clear authority from the start.	Team Lead / Sponsor	Early Project Phase
25	Errors or overly optimistic figures used in budget forecasts.	Risk Reduction	Conduct independent budget reviews and sensitivity testing before baseline approval.	Financial Analyst / Project Manager	Planning Phase

Table 3: Prioritised Risk Register

6.2.2 Updated Risk Register

Rank	Risk	Effect	Probability	Impact	Risk Value	Affected Field(s)
1	Supplier delays in delivering automation hardware	Slows down system integration, pushing back the warehouse launch.	4	5	20	Procurement
2	Software integration failures	Leads to system downtime and inefficiencies in operations.	3	5	15	Engineering
3	Cybersecurity breach	Puts sensitive data at risk, causing financial losses and reputational harm.	2	5	10	Engineering Quality
4	Warehouse automation system failure	Can bring warehouse operations to a halt, leading to financial setbacks.	3	5	15	Engineering
5	High employee resistance to automation	Possible strikes or protests could slow down implementation.	4	3	12	Project Management
6	Poor supplier quality for warehouse components	Increases maintenance costs and reduces system efficiency.	3	4	12	Supplier Quality
7	Budget overruns due to unforeseen costs	Creates financial strain, possibly exceeding the allocated budget.	3	3	9	Procurement
8	Regulatory compliance issues	Could lead to project delays and potential fines.	2	4	8	Project Management

9	Equipment shortages during system implementation	Slows down project progress and raises costs. Increased costs	2	5	10	Manufacturing
10	Unrealistic automation expectations from stakeholders	Can result in disappointment and loss of confidence in the project.	2	3	6	Commercialization
11	Employees Absence	Causes downtime and a drop in overall efficiency.	2	5	10	Engineering
12	Incomplete risk assessment in early stages	Could result in unforeseen project risks down the line.	3	4	12	Market Assessment
13	Delays in milestone approvals	Disrupts the project timeline and affects deadlines.	3	5	15	Project Management
14	Design flaws in automation system	Leads to inefficiencies and unexpected technical challenges.	2	3	6	Design
15	Failure in automated order fulfilment process	Slows down the order processing workflow, affecting deliveries.	3	5	15	Manufacturing
16	Unexpected maintenance costs for automated systems	Adds financial pressure due to unforeseen repair and maintenance needs.	2	4	8	Engineering Quality
17	Organisational training directive	Reduces project budget and causes resource downtime.	2	3	6	HR / Project Budget
18	Top-down budget cuts	Causes significant financial strain mid-project.	3	5	15	Project Budget / Management
19	Community unrest	Disrupts staff availability and project schedule.	2	4	8	Operations / HR
20	Environmental compliance failure	Results in legal penalties and budget overrun.	2	5	10	Engineering / Environmental Compliance
21	Critical resource shortage	Delays in technical tasks due to unavailable specialist skills.	3	4	12	Project Resources
22	Lack of dedicated project manager	Leads to coordination issues and miscommunication.	3	4	12	Project Management
23	No marketing representative	Public relations issues not addressed proactively.	2	3	6	Stakeholder Relations
24	Overloading resources for performance stats	Leads to burnout and inefficiencies.	3	3	9	Resource Management
25	Inaccurate budget assumptions	Causes planning discrepancies and funding shortfalls.	3	4	12	Finance / Project Management
26	Use of tasks to boost resource stats	Introduces inefficiencies and distracts from high-impact work.	3	3	9	Project Operations

Table 4: Updated Risk Register 1

Rank	Cause	Strategy	Prevention Response	Risk Owner	Date Expected
1	Global supply chain problems and unreliable vendors slow down deliveries.	Risk Mitigation	To avoid delays, multiple suppliers will be contracted, and backup stock will be kept on hand.	Industrial Engineer	16 Sep 2025 - 06 Jan 2026
2	The new automation tools might not be fully compatible with Takealot's current system.	Risk Reduction	The system will undergo thorough pre-launch testing, and deployment will happen in phases to catch and fix issues early.	Data Engineer	11 Aug 2025 - 18 Dec 2025
3	Weak security measures make the system vulnerable to hacks and data leaks.	Risk Avoidance	Strong cybersecurity measures will be put in place, including frequent security audits to protect systems from breaches.	Data Engineer	Ongoing, critical during system launch (Feb 2026)
4	Hardware failures, software bugs, or unexpected malfunctions disrupt operations.	Risk Reduction	Scheduled maintenance will help prevent failures, and manual backup operations will be ready as a safety net.	Mechatronic Engineer	06 Nov 2025 - 10 Feb 2026
5	Fear of losing jobs, lack of training, and uncertainty about new technology.	Risk Reduction	Employees will be trained on the new system, and automation will be introduced gradually to ease the transition.	Project Manager	28 Feb 2025 - 13 Feb 2026
6	Some vendors may not meet quality standards, leading to defective or unreliable components.	Risk Control	Vendors will be carefully vetted, and sample testing will be conducted before approving any suppliers.	Quality Engineer	17 Oct 2025 - 23 Dec 2025
7	Extra costs for training, buying hardware, and system integration add up.	Contingency Planning	A financial reserve will be maintained to cover unforeseen costs, and spending will be monitored closely.	Project Manager	28 Feb 2025 - 13 Feb 2026
8	New government rules or safety standards might not be met, causing delays or fines.	Risk Avoidance	Legal experts will be involved early to ensure all industry and safety regulations are met.	Quality Engineer/ Project Manager	Ongoing, critical in commercialization (Feb 2026)
9	Key automation tools, like robotic systems, may be hard to find or take too long to arrive.	Risk Transfer	Early purchase agreements will be secured, and supplier options will be diversified to minimise risk.	Operations Specialist	06 Nov 2025 - 10 Feb 2026
10	Overconfidence in automation's capabilities can lead to disappointment when results take longer than expected.	Risk Acceptance	Clear KPIs will be set, and stakeholder education sessions will be conducted to manage expectations.	Marketing Manager	Pre-launch Phase: May 2025 - Feb 2026
11	Unplanned leave, illness, or other unforeseen reasons reduce productivity.	Risk Reduction	Cross-training programs will be implemented so employees can cover multiple roles if needed.	Project Manager	Aug 2025 - Jan 2026

12	Lack of thorough risk assessment in the early stages leads to surprises later.	Risk Control	AI-driven forecasting and flexible order fulfilment strategies will be used to adjust plans based on market conditions.	Marketing Manager	28 Feb 2025- 12 May 2025
13	Slow decision-making and approval processes push back important deadlines.	Risk Reduction	Regular project reviews and risk updates will be conducted to prevent approval bottlenecks.	Project Manager	28 Feb 2025 - 13 Feb 2026
14	Errors in the planning or implementation phase lead to inefficiencies.	Risk Avoidance	Digital simulations will be used to test automation designs before deployment, with real-time feedback incorporated into improvements.	Product Designer	12 May 2025- 11 August 2025
15	Technical issues slow down or disrupt order fulfilment, causing delays.	Risk Mitigation	Backup procedures will be established, including manual overrides and stress testing of automation workflows.	Operations Specialist	11 June 2025 - 10 Feb 2026
16	Equipment failures and unplanned repairs create additional expenses.	Contingency Planning	A budget will be set aside for maintenance, and predictive maintenance tools will be used to reduce surprise breakdowns.	Project Manager	16 Sep 2025- 23 December 2025
17	Company-wide mandatory training imposed without warning.	Contingency Planning	Include buffer budget and schedule flexibility for unexpected corporate directives.	Project Manager	Period 2
18	Executive-level decision to reduce project funding.	Contingency Planning	Expand contingency reserves; include early-warning process for exec decisions.	Project Manager	Period 5
19	Unrest in surrounding area causing safety concerns and absence.	Risk Transfer	Consider remote work alternatives and include force majeure clauses.	Operations Manager	Period 4
20	Operational emissions exceeded permitted levels during testing.	Risk Avoidance	Run environmental checks and audits before production begins.	Quality Engineer	Period 9
21	Key personnel not available when workload peaks.	Risk Reduction	Secure key skills early; enable cross-training for key tasks.	Project Manager	Period 8
22	Responsibility shared across roles without a central lead.	Risk Control	Assign a project manager with clear authority from the start.	Team Lead / Sponsor	Early Project Phase
23	Marketing role left unassigned during project planning.	Risk Avoidance	Ensure all key functional roles are filled from the outset.	HR / Team Lead	Early Project Phase
24	Over-allocation driven by performance metrics.	Risk Reduction	Distribute work evenly and monitor utilisation carefully.	Project Manager	Ongoing
25	Errors or overly optimistic figures used in budget forecasts.	Risk Reduction	Conduct independent budget reviews and sensitivity testing before baseline approval.	Financial Analyst / Project Manager	Planning Phase
26	Tasks assigned for performance metrics rather than value.	Risk Control	Prioritise tasks with clear value contribution and review assignments regularly.	Project Manager	Throughout Project

Table 5: Updated Risk Register 2

6.3 Appendix C: EVA Calculations for period 9

Period	EV	%COMPLETE	PV	% SCHEDULED	AC	Cum Baseline Cost	Baseline Cost	SPI	CPI
1	\$ 52,934.00	100%	\$ 52,934.00	100%	\$ 78,249.56	\$ 51,684.00	\$ 51,684.00	1.000	0.676
2	\$ 42,454.80	100%	\$ 42,454.80	100%	\$ 52,488.86	\$ 92,888.80	\$ 41,204.80	1.000	0.730
3	\$ 50,139.60	100%	\$ 50,139.60	100%	\$ 69,044.25	\$ 141,778.40	\$ 48,889.60	1.000	0.728
4	\$ 22,446.00	100%	\$ 22,446.00	100%	\$ 20,422.29	\$ 162,974.40	\$ 21,196.00	1.000	0.763
5	\$ 28,641.20	100%	\$ 28,641.20	100%	\$ 20,797.98	\$ 190,365.60	\$ 27,391.20	1.000	0.816
6	\$ 29,354.00	100%	\$ 29,354.00	100%	\$ 27,947.84	\$ 218,469.60	\$ 28,104.00	1.000	0.840
7	\$ 18,978.00	100%	\$ 18,978.00	100%	\$ 30,653.18	\$ 236,197.60	\$ 17,728.00	1.000	0.818
8	\$ 27,681.20	100%	\$ 27,681.20	100%	\$ 23,405.43	\$ 262,628.80	\$ 26,431.20	1.000	0.844
9	\$ 7,213.27	61%	\$ 10,912.00	100%	\$ 25,299.87	\$ 273,540.80	\$ 10,912.00	0.987	0.803
10	\$ -	0%	\$ -	0%	\$ -	\$ 293,188.80	\$ 19,648.00	0.000	0.000
11	\$ -	0%	\$ -	0%	\$ -	\$ 321,408.80	\$ 28,220.00	0.000	0.000
12	\$ -	0%	\$ -	0%	\$ -	\$ 336,048.80	\$ 14,640.00	0.000	0.000
TOTALS	\$ 279,842.07	72%	\$ 283,540.80	75%	\$ 348,309.25		\$ 336,048.80		

Table 6: Period 9, 4 days before

ANALYSIS FOR PERIOD 9		Calculations for the respective answers:	Total Baseline Cost with Milestones	\$ 346,048.80
PV COST	\$ 283,540.80	Total PV (D14)		
AC	\$ 348,309.25	Total AC (F14)		
EV COST	\$ 279,842.07	Total EV (B14)		
BAC	\$ 336,048.80	Total Baseline Costs (K14)		
CPI	0.803	Total EV Cost / Total AC		
SPI	0.987	Total EV Cost / Total PV Cost		
ETC	\$ 69,958.47	(BAC-Total EV Cost) / CPI		
EAC	\$ 418,267.71	Total AC+ ETC		
Project is	Behind Schedule			
Project is	Over Budget			

Figure 7: period 9 analysis with respective calculations and answers

Period	EV	PV	AC	SPI	CPI	ETC	EAC	Cum Baseline Cost	Baseline Cost
1	\$ 49,464.04	\$ 50,046.27	\$ 78,249.56	0.988	0.632	\$ 469,181.74	\$ 547,431.30	\$ 51,684.00	\$ 51,684.00
2	\$ 92,644.41	\$ 91,154.24	\$ 130,738.42	1.016	0.709	\$ 357,600.51	\$ 488,338.93	\$ 92,888.80	\$ 41,204.80
3	\$ 142,713.29	\$ 142,542.67	\$ 199,782.67	1.001	0.714	\$ 284,647.01	\$ 484,429.67	\$ 141,778.40	\$ 48,889.60
4	\$ 162,830.64	\$ 164,078.83	\$ 220,204.96	0.992	0.739	\$ 247,776.15	\$ 467,981.10	\$ 162,974.40	\$ 21,196.00
5	\$ 194,460.56	\$ 194,393.67	\$ 241,002.94	1.000	0.807	\$ 187,869.52	\$ 428,872.46	\$ 190,365.60	\$ 27,391.20
6	\$ 225,257.27	\$ 225,969.60	\$ 268,950.78	0.997	0.838	\$ 144,221.66	\$ 413,172.44	\$ 218,469.60	\$ 28,104.00
7	\$ 242,371.95	\$ 244,947.60	\$ 299,603.95	0.989	0.809	\$ 128,158.38	\$ 427,762.33	\$ 236,197.60	\$ 17,728.00
8	\$ 269,499.83	\$ 272,628.80	\$ 323,009.38	0.989	0.834	\$ 91,747.87	\$ 414,757.25	\$ 262,628.80	\$ 26,431.20

9	\$ 279,842.07	\$ 283,540.80	\$ 348,309.25	0.987	0.803	\$ 82,405.10	\$ 430,714.35	\$ 273,540.80	\$ 10,912.00
10	\$ 298,892.34	\$ 303,188.80	\$ 366,204.25	0.986	0.816	\$ 57,776.30	\$ 423,980.55	\$ 293,188.80	\$ 19,648.00
11	\$ 328,704.89	\$ 331,408.80	\$ 389,089.16	0.992	0.845	\$ 20,530.04	\$ 409,619.21	\$ 321,408.80	\$ 28,220.00
12	\$ 346,048.80	\$ 346,048.80	\$ 403,239.45	1.000	0.858	\$ (0.00)	\$ 403,239.45	\$ 336,048.80	\$ 14,640.00

Table 7: Each period analysed 4 days before

6.4 Appendix D: Resource Development

Training Decisions for K1 in period 1

View Project Plan

Export to XLS

Export to PDF

Drag a column header here to group by that column

Resource	Resource Category	Training	Duration (Days)	Cost	Lost Work	Total
Marthe Huber	Operations Specialist	Interpersonal skills	3	\$600	\$1,272	\$1,872
Marthe Huber	Operations Specialist	Principles of Quality	3	\$600	\$1,272	\$1,872
Valene Stimson	Junior Product Designer	Process Engineering	3	\$600	\$1,152	\$1,752

Lost Work = (# of Days Training) * (8 Hour/Day) * (Resource Rate \$/Hour)

Training Decisions for K1 in period 2

View Project Plan

Export to XLS

Export to PDF

Drag a column header here to group by that column

Resource	Resource Category	Training	Duration (Days)	Cost	Lost Work	Total
Kong Duanmù	Engineer	Project Management 101	5	\$1,000	\$2,000	\$3,000

Training Decisions for K1 in period 3

View Project Plan

Export to XLS

Export to PDF

Drag a column header here to group by that column

Resource	Resource Category	Training	Duration (Days)	Cost	Lost Work	Total
Marthe Huber	Operations Specialist	Process Engineering	3	\$600	\$1,272	\$1,872

Figure 8: Resource training for the simulation

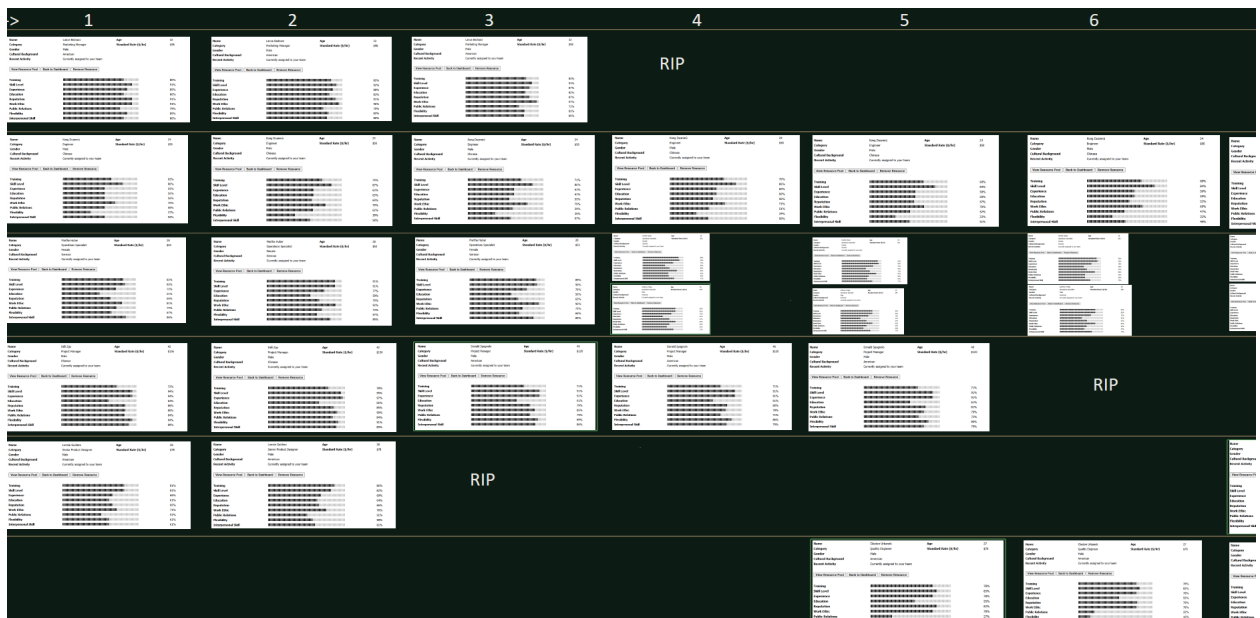


Figure 9: Resource progression per round

6.5 Appendix E: Meeting Minutes



GROUP K1

PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 1

THURSDAY 12:00, 06 MARCH 2025

CIVIL ENGINEERING SS CUBICLES

Minutes

Call to order		12:02
Attendance	Taro	All members present.
Resource Allocation and Bidding Strategy		
	Flippie	• Suggested adding an extra team member. • Proposed bidding slightly higher (up to 10 above baseline) to secure key resources.
	Lauren	• Concerned other groups will bid significantly higher; currently only secured one resource at base salary.
	Flippie	• Noted simulation resets tomorrow, suggested experimenting with bold strategies tonight. • Recommended a possible experiment involving firing and rehiring resources. • Proposed intentionally not bidding to lower other groups' bids, making future rounds cheaper.
	Lauren	• Recommended training someone specifically for design tasks. • Highlighted that training counts as unproductive time, incurring extra costs (~\$1000).



	Taro	Warned that experimental strategies tonight could backfire.
	Emmanuel	Mentioned some resources already assigned tasks.
Task Assignments		
	Lauren	Agreed on assigning two people to the advertising campaign due to high hour requirements.
	Luke	Confirmed Lance is assigned 100% and Max at 50% after first period
	Emmanuel	- Noted advertising campaign duration is long; expressed desire to finish early to move onto other tasks. - Will share updated MS Project file for team collaboration.
Training and Resource Development		
	Lauren	Suggested marketing training for team members
	Flippie	- Recommended experimenting tonight by training multiple team members. - Created a shared document repository on MS Teams.
	Lauren	Confirmed the Quality Engineer now possesses project management skills after training.
	Emmanuel	- Highlighted that mandatory training affects budgeting significantly. - Assigned Ronald Wachtel (high work ethic post-training) heavily in Project Management (60%).

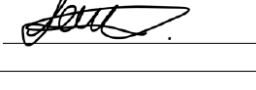


	Lauren	Interested to observe how Ronald's low flexibility impacts performance.
	Luke	Asked about improving employee flexibility through salary increases.
	Emmanuel	Clarified that salary adjustments require firing and rebidding.
Workload and Efficiency		
	Taro	Emphasized avoiding overtime; uncertain about overtime's impact on overall performance.
	Lauren	Pointed out Lonnie's current 160% assignment.
	Emmanuel	Explained some seemingly over-assigned resources are sequential tasks, thus not truly overburdened.
Staffing Strategy and Team Cohesion		
	Flippie	- Recommended initially hiring specialists and training junior staff to eventually replace senior team members. - Junior staff cost significantly less; suggested pairing juniors with seniors for effective training.
	Luke	Suggested juniors might gain experience faster when working with senior colleagues.
	Emmanuel	Advised caution as frequent team changes can negatively impact team cohesion.



	Taro	Recommended balancing junior/senior staff to maintain cohesion while building skill levels.
Project Management and Budget Control		
	Emmanuel	Stressed importance of budget consideration in every decision.
	Flippie	Raised concern about funding sources.
	Taro	Clarified there's no additional income; budgeting must be careful to avoid shortages at the project end.
	Lauren	- Will confirm additional income with professor. - Explained that new tasks become available only after current tasks are completed in each period.
	Emmanuel	Recommended aiming for 100% task completion without employee burnout.
	Taro	Clarified additional staff won't necessarily linearly increase productivity.
	Lauren	Suggested checking task dependencies to improve efficiency by assigning related tasks to the same resources.
Employee Engagement and Team Longevity		
	Flippie	- Planned managerial actions (e.g., pizza party) to test impacts on team metrics. - Experimenting by inviting only half the team to an office party to gauge metric changes.



	Lauren	Scheduled Martha for a three-day Quality Principles training.
		Considered firing Martha post-training, noting improved skills without a salary increase.
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		13:19

Addendums

- Sim4Projects Resource Allocation
- K1 WhatsApp Group for bidding details



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 2
MONDAY 13:00, 10 MARCH 2025
CIVIL ENGINEERING SS CUBICLES

Minutes

Call to order		13:00
Attendance	Taro	<i>All members present.</i>
Assignment 2 Overview		
	Lauren	• Created a template for Assignment 2. • Suggests working in pairs on different sections. • Deliverables include: ◦ PDF Report ◦ MS Project file ◦ Meeting minutes • Must decide the final product considering available resources and tasks. • Report needs to be detailed, concise, and budget focused. • Risk management is crucial, with an upcoming lecture dedicated to this. • Must choose between prioritizing responsiveness (speed) or cost-effectiveness. • Highlighted prize incentive for the team achieving the lowest cost and fastest completion.
	Flippie	• Suggested balancing responsiveness and cost-effectiveness.



	Lauren	Noted difficulty in balancing both; advised prioritizing clearly.
	Emmanuel	Mentioned that the trial run indicated the project would only last six weeks.
	Lauren	- Warned about limited opportunities for additional funding. - Recommended the main body of the report to be no more than 18 pages, clearly sectioned (e.g., Project Scope).
Project Timeline		
	Taro	Suggested targeting project completion by Friday.
	Lauren	Noted assignment instructions allow significant flexibility.
Budget and Resource Allocation		
	Emmanuel	Indicated trial run exceeded budgets, especially due to overtime work.
	Lauren	- Recommended clearly defining resources to avoid cost overruns. - Emphasized importance of detailed minutes.
	Emmanuel	Requested clarity on baseline functionality in MS Project.
	Lauren	- Explained baseline feature (initial resource allocation plan for future comparison). - Recommended reading further clarifications provided on the forum.
	Flippie	Suggested clarifying details regarding baseline and resource specifics through the forum.

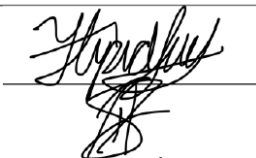
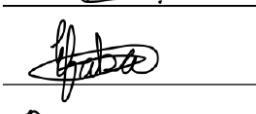
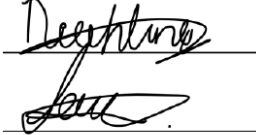
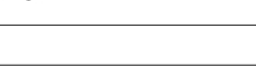
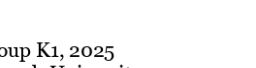


	Taro	- Recommended assigning tasks immediately, then discussing further details in the group chat. - Suggested completing a first draft by Thursday's meeting.
Task Assignments		
		- Executive Summary: Taro and Luke - Project Scope: Taro and Luke - Baseline Project Plan: Emmanuel and Flippie - Project Budget: Lauren (will request assistance if needed) - Risk Management Plan: Lauren (final review in group meeting)
	Emmanuel	Recommended separate documents for each section.
	Luke	Mentioned lack of marking sheets, advised relying on tutorial session documents.
Resource Allocation Planning		
	Lauren	- Needs to finalize Gantt chart. - Unsure about baseline updates between MS Project versions.
	Flippie	Prioritize allocating experienced and junior staff appropriately.
	Lauren	Advised considering task dependencies and assigning related tasks to the same resources.
	Luke	Advised pre-planning wage allocations carefully.



	Emmanuel	Recommended utilizing baseline views in Gantt charts for insights.
	Lauren	Warned against resource overallocation, as it typically causes delays.
	Luke	Recommended incorporating safety margins in allocations.
	Taro	Suggested allocating only 95% capacity to ensure timely task completion.
	Emmanuel	Raised concern regarding poor budget breakdown and cost variance.
	Lauren	Clarified sequential allocations; tasks remain proportionally allocated even after preceding tasks finish.
Staffing and Hiring Concerns		
	Luke	Noted understaffing in trial run; suggested hiring approximately 10 employees eventually.
	Taro	Recommended reviewing resource pool post-first bid round and reconsidering approach thereafter.
	Flippie	Mentioned having three initial bids; suggested focusing primarily on immediate needs.
		- Project Manager: Delhi (bid: \$136) - Marketing Manager: Lance (bid: \$100) - Junior Marketing: Max (bid: \$57) - Engineer: Cody (bid: \$51) - Senior Product Designer: Lonnie (bid: \$80) - Junior Product Designer: Valene (bid: \$48)



		- Operations Specialist: Marthe (bid: \$53) - Agreed to delay hiring a quality engineer for now.
	Emmanuel	Noted most groups will likely stick with their trial-run hires.
	Taro	Believes most bids will be lower compared to the trial.
Task Prioritisation and Risk Management		
	Emmanuel	Raised question on prioritizing critical vs. non-critical tasks.
	Luke	Asked about methods for risk estimation.
	Taro	Recommended deferring final decisions until Thursday's risk lecture.
		- Action Item: All members should share product ideas in the group chat. - Product Choice: Smart Warehouse Management System and Optimisation
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		14:31



Addendums

- Sim4Project Bids and Resources
- Assignment 2 Document on STEMLearn



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 3
Thursday 13:10, 14 MARCH 2025
EENDRAG HK-KANTOOR

Minutes

Call to order		
Attendance	Taro	<i>All members present.</i>
General Matters		
	Taro	• Minutes will be finalised by tomorrow afternoon. • Milestones should be extracted from MS Project.
	Lauren	• Milestones are dependent on major events from the overall project. • Suggested reviewing Gantt chart to identify major events and completions. • Will create Work Breakdown Structure (WBS) for each period, detailing major events. • Will update team if there are adjustments to scheduling.
	Emmanuel	• Agreed to insert and update information in MS Project as needed.
Budget and Resource Allocation		
	Lauren	• Budget is almost fully allocated. • Approximately \$15,000 currently set aside as contingency but ideally needs more.



		Professor recommended avoiding hiring a dedicated project manager; engineers could manage tasks effectively.
	Luke	Will integrate the budget and Work Breakdown Structure (WBS) with project scope.
	Lauren	Will finalise the risk management document categorising risks into High, Medium, and Low for team review.
	Flippie	- Suggested hiring another engineer as soon as one becomes available instead of a project manager. - Emphasised that engineers are competent for most tasks.
	Lauren	- Recommended hiring a quality engineer who would be more cost-effective. - Mentioned having an additional "hidden" contingency but would prefer not to utilize it.
Staffing Concerns		
	Flippie	Anticipates heavy workload over the next two periods.
	Taro	Raised that the marketing team can be reduced after the marketing-intensive periods have passed, freeing up some budget
	Lauren	- Noted difficulty budgeting for unpredictable marketing issues. - To manage budget constraints, recommended firing the current project manager and hiring an additional engineer before Period Four. - Also noted marketing manager costs seem high.

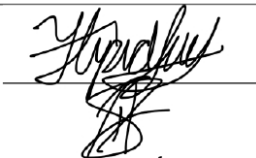
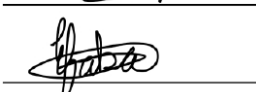
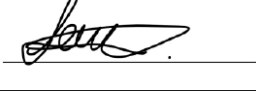


	Flippie	- Recommended retaining the marketing manager for the current period. - Suggested operations specialist could support project management tasks alongside engineers.
	Emmanuel	- Agreed existing resources have the capacity for project management. - Recommended a balanced approach, allocating project management tasks with a ratio of 70% engineer and 30% project manager.
	Lauren	Noted this approach would raise contingency funds to around \$19,000.
	Emmanuel	Pointed out savings achieved by sending unallocated resources to training sessions (which is less costly than unallocated hours).
	Flippie	Highlighted benefits of training a junior marketing specialist.
	Emmanuel	Recommended training an engineer in project management, though benefits of training will only be apparent by Period 2 or 3.
	Flippie	Proposed part-time training for the engineer alongside project management tasks.
Resource Management		
	Luke	- Emphasized caution in assigning tasks outside team members' primary roles to avoid delays. - Questioned average personnel allocation per task.



	Lauren	- Clarified most tasks typically have one or two people allocated based on required skills and time. - Noted professor advised against over-allocating resources.
	Flippie	Recommended keeping an eye on the resource pool for additional engineers to hire for future periods.
	Emmanuel	Highlighted the need to plan for hiring another specialist towards the end of the period.
	Flippie	Suggested exploring optimal junior-to-senior resource allocation ratios.
	Emmanuel	Indicated the current balance in resource allocation is generally effective.
Specific Issues		
	Lauren	Raised concern about the high expense of using Delhi (currently hired project manager); recommended assigning tasks to engineers instead, significantly reducing costs.
	Flippie	Proposed temporarily keeping the project manager but significantly reducing hours until another engineer is hired.
	Lauren	Highlighted a budgeting issue: unable to budget for an engineer who hasn't yet been hired.
Baseline and Documentation		
	Emmanuel	Wrote a paragraph outlining the approach for the baseline (available on Teams).



		- Mentioned resource allocation centred on maintaining timelines for critical paths. - Included notes on a 2% penalty and provisions for gaps.
	Lauren	Recommended adding a network diagram to enhance the thoroughness of the documentation.
Upcoming Duties		
	Luke	Work on milestones table for project scope.
	Lauren	Finalize risk document, update budget, and complete WBS.
	Emmanuel	Adjust MS Project file, confirm resource allocations, finalize baseline plan based on new changes.
	Flippie	Work on project baseline and corresponding report.
	Taro	Finalize and edit project scope, finalize meeting minutes.
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment	14:25	



Addendums

- Budget
- Baseline Project Plan
- Work Breakdown Structure
- Milestones
- Project Scope



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT RISK MANAGEMENT MEETING
SUNDAY 15:00, 16 MARCH 2025
MS TEAMS

Minutes

Call to order		15:02
Attendance	Taro	<i>All members present</i>
Objectives		
	Lauren	• Finalize project risk probabilities, impacts, and assigned responsibilities. • Ensure clarity and relevance of identified risks. • Risk Levels defined: ◦ High: >14 ◦ Medium: 7–14 ◦ Low: <7
Detailed Risk Review		
Supplier Delays	Risk Level: 20, High	• Cause: Global supply chain disruptions, vendor inefficiencies. • Effect: Delay in system integration and warehouse launch. • Strategy: Risk mitigation (multiple suppliers, emergency stock).
Software Integration Failures	Risk Level: 15, High	• Cause: Compatibility issues with existing systems. • Effect: Downtime, reduced operational efficiency. • Strategy: Risk reduction (pre-launch testing, phased development).



Cybersecurity Breach	Risk Level: 10, Medium	• Cause: Weak security protocols, external hacking attempts. • Effect: Data leaks, financial losses, reputational damage. • Strategy: Risk avoidance (enhanced security measures, audits). • Action: Operations Specialist assigned. Consider separate mention of broader organizational risks (e.g., Takealot).
Warehouse Automation System Failure	Risk Level: 15, High	• Cause: Hardware malfunctions, software bugs. • Effect: Operational halt, financial losses. • Strategy: Risk reduction (scheduled maintenance, manual backups).
High Employee Resistance to Automation	Risk Level: 12, Medium	• Cause: Fear of job loss, insufficient training. • Effect: Reduced productivity, morale issues, potential strikes. • Strategy: Employee training, gradual rollout.
Poor Supply Quality	Risk Level: 12, Medium	• Cause: Unverified vendors, inadequate quality control. • Effect: Increased maintenance, reduced efficiency. • Strategy: Strict vendor qualification, sample testing.
Budget Overruns	Risk Level: 9, Medium	• Cause: Unforeseen training, hardware, and integration expenses. • Effect: Financial strain, exceeding budget. • Strategy: Contingency planning, financial reserves, expense monitoring.

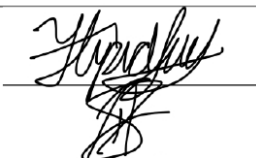
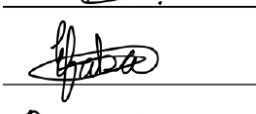
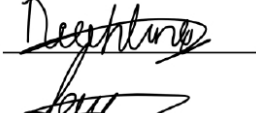
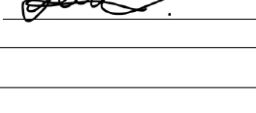
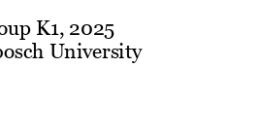


Regulatory and Compliance Issues	Risk Level: 8, Medium	• Cause: Changing government policies, non-compliance. • Effect: Project delays, fines. • Strategy: Early engagement of legal experts, compliance with standards.
Equipment Shortages	Risk Level: 10, Medium	• Cause: Unavailability of critical systems. • Effect: Delays, increased costs. • Strategy: Early procurement agreements, diversified suppliers.
Unrealistic Stakeholder Expectations	Risk Level: 6, Low	• Cause: Overestimation of capabilities. • Effect: Stakeholder disappointment, confidence loss. • Strategy: Setting realistic KPIs, stakeholder education.
Employee Absenteeism	Risk Level: 10, Medium	• Cause: Unplanned absence (illness, emergencies) of skilled workers. • Effect: Reduced efficiency, task delays. • Action: Add measures for critical absenteeism management, train more than one resource for specific critical tasks
Additional Risk Discussions		
	Taro	• Recommended clearly assigning risk ownership by specific technical areas (data, mechatronics, etc.).
	Lauren	• Clarified project team representation differs from Sim4Projects; current team has five engineers.
	Luke	• Suggested integrating closely related risks (#1 and #6).
	Lauren	• Requested members review the final compiled risk document thoroughly.



	Luke	Clarified risk #5 regarding employee shifts versus terminations: employees worried about both, risk should specify this.
	Lauren	Noted an unexpected salary expense issue due to a Sim4Projects training glitch; suggested caution about training timing.
	Flippie	Expressed confidence despite initial setbacks; advised reviewing planned budget allocations.
	Emmanuel	Highlighted cost savings through efficient engineer utilization.
	Lauren	Recommended retaining historical data (such as Sim4Projects and MS Project statistics) for future reference, and for Deliverable 3.
Project Management		
	Lauren	Consider training an engineer in project management to replace the expensive project manager role.
	Flippie	Suggested a reduced bid (\$120) for a Project Manager role in Period 3.
	Emmanuel	Agreed to train engineers for project management to minimize financial burden.
	Taro	Believed initial extra costs could balance out with a focus on fundamentals in the next period.
	Emmanuel	Raised concerns about juniors' performance, proposed balanced assignment (Project Manager 40%, Engineer 80% while training).



Resource Allocation for Upcoming Period		
		Design/Development: Valene (100%), Lonnie (40%)
		Marketing Program: Max (80%), Lance (50%)
		Preliminary Marketing: Max (100%)
Tasks to complete before Deliverable 2 submission		
	Luke & Taro	Project document conclusion and executive summary.
	Lauren	Finalise risk register, complete main submission document formatting.
	Flippie	Complete network diagram and review MS Project File.
	Taro	Finalise meeting minutes.
	Emmanuel	Complete MS Project File (team to review).
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		16:36



Addendums

- Final Budget (Risk Register)
- Final Budget (Resource Cost Sheet)
- Group K1 Deliverable 2 Document
- Sim4Projects Allocations for Period 2



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 5
TUESDAY 16:00, 08 APRIL 2025
MS TEAMS

Minutes

Call to order		16:03
Attendance	Taro	<i>All members present</i>
Budget and Resource Overview		
	Lauren	- Noted that the team is currently in the red overall. Although Period 4 saw minor savings, significant losses were accumulated in Periods 1–3. Suggested focusing on cost-saving strategies moving forward. - Confirmed that Max and Lonnie were fired in the previous period, but the team did not secure a cheaper engineer through the bid. The same engineer is still onboard.
Role Assignments and Task Allocations		
	Emmanuel	- Announced that an Operations Specialist was hired to help with Project Management. - Proposed allocating more of the Project Management task to the Ops Specialist to save costs.



		• Mentioned that there's slack in project management, so minor delays wouldn't be disastrous.
	Flippie	• Observed that Martha and Anthony are likely not needed in the next few tasks. • Confirmed plan to use them for management tasks instead. • Recommended bidding for two Quality Engineers. • Noted that no new engineers are available for cheaper bids.
	Taro	• Asked about the availability of cheaper engineers. • Suggested a bidding strategy: \$70 for Clayton (first choice) and \$69 for Urith (second choice), balancing cost and likelihood of success. • Proposed re-evaluating efficiency of past allocations to guide future assignments.
	Lauren	• Confirmed Valene is no longer necessary and should be fired. • Noted inefficiency when two people are placed on the same task—prefer single-person assignment where possible. • Proposed 100% assignment of engineer to engineering tasks.
	Lukas	• Asked when resources will be needed next for procurement (answer: around Period 11). • Raised point that engineering tasks are on the critical path—resource decisions here are crucial.

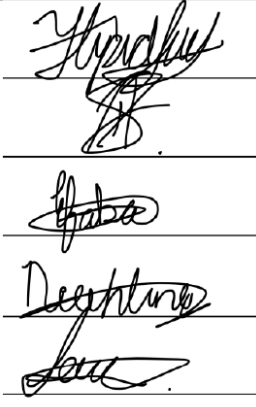


		• Suggested a 40% allocation for the Project Manager to save costs. • Recommended monitoring Quality Engineer performance if hired, and noted they could help with build on the critical path.
	Taro	• Asked whether there will be training this period—confirmed there will not be. • Proposed smaller-scale celebration or reward for morale—such as a pizza party or recognition for most improved team members.
	Lauren	• Deemed a milestone celebration too expensive for this period. • Suggested delaying it to the next period after cost-saving measures are in place. • Proposed increasing Martha's allocation to 30% for pre-production, a long engineering task.
	Emmanuel	• Suggested reducing Operations Specialist allocation from 30% to 20% if budget is a concern. • Flagged that the engineer's efficiency has been dropping (5.75 days vs. 5 days for last task). • Recommended checking if Quality Engineers can support critical build tasks. • Highlighted that both procurement tasks in next period require Operations Specialists and are on the critical path.
	Lukas	• Asked if Ops Specialist tasks should be split or assigned individually.



		• Recommended assigning an engineer to the build function and possibly including a Quality Engineer if available. • Suggested one Ops Specialist at 100% for longer procurement task, and the other at 75%, freeing time to support engineering if needed.
	Lauren	• Noted that even if Ops Specialist tasks take double the time, they won't impact the critical path. • Agreed on 75/75% allocation split for Ops Specialists if needed. • Confirmed that each Quality Engineer, if secured, will be assigned one specific engineering task for optimal efficiency.
Assignment Decisions		
		• Bid Submissions: ◦ \$70 for Clayton ◦ \$69 for Urith (Quality Engineers) • Resource Allocations: ◦ Engineer at 100% for engineering tasks ◦ Project Manager at 40% ◦ Operations Specialist at 80% (Project Management) ◦ Martha at 30% (for pre-production engineering task) • Valene to be fired • No training this period • Postpone celebration or recognition event to next period



		Each task should have only one assignee where possible to maintain efficiency
Efficiency Notes		
		- Engineer's output is slowing—consider monitoring and supporting accordingly. - Multiple assignments to one task lower efficiency. - Engineering and Procurement tasks are on the critical path—must allocate carefully.
Next Meeting		
		- Monday, 14 April 13:00-14:00 - Note: Meeting moved to better align with assignment schedule; preference for in-person attendance.
Member Signatures		
	Flippie van der Spuy	
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		17:04

Addendums

- Master Project Plan
- Project Budget



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 6
MONDAY 13:00, 14 APRIL 2025
E1029

Minutes

Call to order		13:05
Attendance	Taro	<i>All members present</i>
Previous Period Review		
	Lauren	• Noted that teams collaborated well during this period. • Financial status is still tight; although things improved slightly, the simulation deducted \$20,000 from every team. • Confirmed that the upcoming task set contains several parallel-running tasks. • Noted that "Approve Sample Parts" is currently assigned at 40/60% and "Design Validation" had the Senior Product Designer allocated.
	Taro	• Asked whether the team lost more money since the last period. • Confirmed that next runs are on Monday, Wednesday, and Thursday.
	Lauren	• Clarified the team is now better off but still over budget. • Stated that they have \$15 000 in contingency funds remaining. • Reaffirmed that the focus will be on completing tasks on time.



		Suggested removing Saidee and adding Ryan to help balance task loads.
	Flippie	- Estimated the team would need about \$44 000 more to complete the project. - Asked about resources required for the next period.
	Emmanuel	- Mentioned that the team didn't finish the "Test Prototype" task in the previous round. - Recommended using Saidee to accelerate it, but noted Marthe would be more effective there. - Proposed assigning Marthe to testing and Kqwang to that task at 30% as well. - Noted that Clayton is most efficient for the testing task since it's his specialization. - Stated that Process Engineering is currently a critical task. - Suggested that Kqwang could instead help with Marketing if needed. - Expressed no concern over Ryan's current under allocation, since he's still new and learning.
	Flippie	Supported the idea of leaving Ryan as is; risk of under allocation is minimal.
	Emmanuel	- Added that even if Ryan logs more than 360 hours, it's acceptable as the cost is fixed. - Warned that the engineer's efficiency is still declining—possibly due for a Disciplinary.

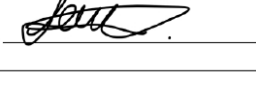


	Lauren	Agreed to wait and see if the engineer improves, else they'll consider reprimanding him next round.
Assignment 3 Planning		
	Lauren	- Stressed the importance of Earned Value Analysis (EVA) for every period, especially Period 9. - Mentioned that managerial costs should be factored into the analysis. - Stated she will begin with cost calculations four days before the deadline and share an updated Excel sheet.
	Taro	- Will upload detailed notes from the tutorial. - Confirmed that the formulas for EVA remain the same. - Mentioned need for meeting minutes from Assignment 2 and insights from Risk Meeting 2 to check which risks have materialised.
	Emmanuel	Highlighted the need to account for baseline changes, but confirmed that the team should still use the original baseline for EVA.
	Flippie	Will create a shared Word document for collaborative writing and allow task delegation next week.
	Lauren	- Confirmed the baseline from Assignment 2 should be included for reference. - Mentioned the tedious nature of SPI calculations and advocated for a



		reusable Excel template across all periods. Will update the risk table and suggested that Risk Realisation delegation happen during Risk Meeting 2.
	Luke	Agreed that SPI calculations will be time-consuming, and that tasks should be divided evenly once all formulas are finished and checked.
Decisions & Action Items		
		- Remove Saidee, add Ryan to task allocation. - Assign Marthe to "Test Prototype". - Assign Kqwang at 30% to assist or reallocate to Marketing. - Keep Clayton on testing due to field expertise. - Monitor engineer's efficiency; possibly reprimand next round. - Prioritise on-time task completion due to tight budget. - Shared EVA responsibilities and document preparation: - Taro: Upload tutorial notes and Assignment 2 minutes - Lauren: Share updated Excel with cost formulas - Flippie: Set up shared Word doc for Assignment 3 - Risk realisation to be finalised in Risk Meeting 2



		○ Include Assignment 2 baseline in new analysis • Risk Meeting 2: Time TBD — includes risk realisation and updated assessments • Assignment 3 work continues throughout the week • Tasks for Assignment 3 are delegated on the Project Document.
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment	13:50	

Addendums

- Project Document Assignment 3
- MS Project File



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 7
TUESDAY 15:30, 22 APRIL 2025
E1029

Minutes

Call to order		15:30
Attendance	Taro	<i>All members present</i>
Sim4Projects Planning		
	Lauren	Confirmed that the team will be getting a \$50,000 boost this period.
	Flippie	- Suggested putting the Quality Engineer onto Procurement. - Asked who should be assigned to Project Management.
	Lauren	Reminded that previously the Engineer and Anthony worked well together on Project Management.
	Flippie	Proposed using Anthony, Marthe, and Clayton for Project Management duties this period.
	Emmanuel	Pointed out that overallocating some of them could help speed things up.
	Lauren	Agreed; suggested each could take on Project Management and one other task.
	Emmanuel	Proposed letting Marthe handle Project Management primarily, since she's already an Ops Specialist for Procurement.

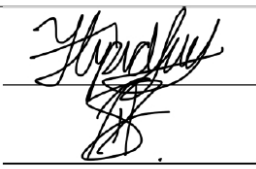
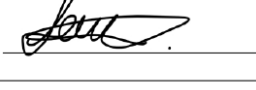
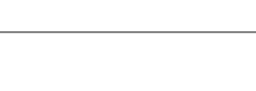


	Flippie	- Recommended allocating 20% Engineer effort to Project Management to help speed up tasks and possibly increasing up to 70%-80% if needed. - Cautioned about not overloading Ops Specialists, as they are crucial for Manufacturing.
	Emmanuel	Noted that Ops Specialists and Quality Engineers work simultaneously, minimizing conflicts.
Resource Management		
	Flippie	- Raised the question of whether they should fire Ryan. - Cited that the next run includes a Product Launch with only 24 hours to complete it. - Suggested possibly bringing in someone smaller (lower cost) for that.
	Lauren	Warned that junior staff are less efficient, so the team would have to compensate carefully to still meet deadlines.
	Emmanuel	Suggested putting Marthe as Project Manager for the Product Launch, given her excellent stats.
	Lauren	Noted they could hire additional people since the final period has six large tasks that must be completed.
	Flippie	Mentioned a Marketing Manager available for \$59 salary.
	Lauren	Proposed underbidding to save funds.
	Flippie	Agreed but cautioned others will likely bid too. Final decision to bid \$56.



	Luke	Highlighted that Felix (another resource) also has good stats.
	Lauren	- Confirmed that Kong will be working 100% across four development tasks. - Confirmed decision to fire Ryan and replace him with the Marketing Manager.
Assignment 3 Progress		
	Lauren	- Mentioned she converted the masterplan into an Excel sheet. - Still working on figuring out date conversions properly.
	Taro	Explained that Sim4Projects imports are mostly manual to account for discrepancies between Sim4Projects and MS Project.
	Lauren	- Acknowledged the lack of clarity around import issues. - Confirmed she will break down the report sections for better task distribution.
	Tasks for Assignment 3	Flippie: Earned Value Analysis (EVA) for Sheets 1-4. Luke: EVA for Sheets 5-8. Lauren: EVA for Sheets 9-12. Emmanuel: Will work on MS Projects and assist with EVA sheets. Taro: Responsible for writing the report.
Decisions & Action Items		
		Fire Ryan and replace with newly hired Marketing Manager (bid \$56).



		• Assign Marthe to Product Management tasks during Product Launch period. • Allocate Engineer partially (20–70%) to assist Project Management. • Keep Ops Specialists mostly on Procurement and Manufacturing critical tasks. • Break down Assignment 3 tasks as per division listed. • Monitor manual vs. automatic updates between Sim4Projects and MS Project. • Next Meeting: Tuesday, 19:00–20:00
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		16:50

Addendums

- Assignment 3 Report
- EVA Sheet
- MSPProject File



GROUP K1
PROJECT MANAGEMENT 412

GROUP PROJECT MEETING 8
TUESDAY 19:00, 29 APRIL 2025
EENDRAG REC HALL

Minutes

Call to order		19:05
Attendance	Taro	<i>All members present</i>
Progress Updates		
	Lauren	• Finished PV, EV, and the Summary Sheet. • Uncertain about how to calculate training costs and managerial actions without period-specific baselines. The class rep has contacted the lecturer for clarification. • Assumes managerial tasks start at the beginning of the period and that bonuses will be awarded accordingly. • Explained how % complete was calculated for EVA. • Asked if the team should create a summary sheet up to Period 12, not just Period 9. • Noted SPI and CPI are already calculated per period, so graphs are not computationally heavy. • Clarified that if referencing data in the report, it should be from the Summary Sheet or Graphs, not raw EVA data. No need to include every period as an appendix.

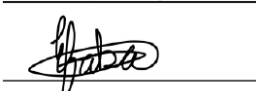
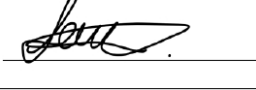


	Emmanuel	• Will write risk plan after Risk Meeting. • Explained the high milestone cost was due to unallocated resources early on. • Uploaded a separate scheduling sheet to Teams. • Pointed out a budget inconsistency from Assignment 2: ◦ Originally had a \$22,000 contingency, ◦ Baseline was \$357,000, ◦ Masterplan said \$346,000, ◦ Final budget became \$403,239.65 with a \$33,951 contingency. • Will complete the EVA summary sheet for all periods. • Also mentioned a risk from overallocating resources.
	Taro	• Asked why a second Risk Meeting was necessary if the register wasn't being expanded. • Will edit other report sections and prepare a template for the shared section. • Suggested that period-specific graphs might be redundant and recommended a second summary sheet instead. • Noted that more information is generally better, even if graphs don't carry specific marks. • Encouraged everyone to contribute to the Lessons Learned and Multidisciplinary Review.



	Luke	• Will update the EVA sheet with totals for each period to ease reading and graphing. • Will add this to Section 2 of the report. • Suggested not expanding the full risk register—it's not a large portion of the spec. • Will explore software to generate graphs. • Tasked with: ◦ Graphs for Period 9, ◦ Graphs for overall period comparison tasks, ◦ Graphs for the EVA summary (including budget/schedule comparison).
Task Assignments		
	Luke	• Create graphs for Period 9. • Create graphs for period tasks and overall EVA summary.
	Emmanuel	• Complete EVA summary sheet for all periods. • Assist with analysis on above/below budget and schedule status for each period.
	Lauren	• Write risk plan following the second risk meeting.
	Taro	• Edit other report sections. • Prepare a template for shared/common section.
	Flippie	• Complete the rest of Section 4.
	Everyone	• Contribute to Lessons Learned and Multidisciplinary Review sections.



		Reflect on personal contributions to multidisciplinary aspects for others to reference.
Risks Identified		
		- No dedicated Project Manager; project management handled by cross-role allocation. - Had to issue a warning to the Engineer to improve performance. - Managerial tasks were used excessively to boost resource stats—proved costly. - Didn't hire a Marketing Manager—a critical gap. - Resource overallocation was used as a risky strategy.
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fako	
	Luke Neethling	
	Lauren Coetser	
Adjournment		21:20

Addendums

- EVA Excel Sheet
- Budget Sheet
- MSProject Data



GROUP K1
PROJECT MANAGEMENT 412

POST COMPLETION PROJECT REVIEW

MONDAY 19:00, 06 MAY 2025
EENDRAG REC HALL

Minutes

Call to order		19:04
Attendance	Taro	<i>All members present</i>
Agenda		
		• Review of final performance metrics (EVA, CPI, SPI) • Integration and analysis of simulation performance • Discussion of team development and resource strategy • Reflections on real-world and simulated project synergy • Lessons learned • Evaluation of multidisciplinary collaboration
EVA Review		
	Lauren	• Presented the compiled EVA spreadsheet, highlighting the trends in Actual Cost vs Budget and Baseline vs Actual Schedule. Noted that: ◦ Early project phases saw time efficiencies but incurred higher costs. ◦ Later phases prioritised cost containment through targeted training and role substitution



		(e.g., using engineers in PM tasks). Overall, the project closed \$8239.65 over budget, which is minimal considering the scope and learning curve.
	Luke	Added that the EVA graphs from Excel clearly support this narrative and will be appended to the closure report.
Budget Analysis		
	Taro	Noted that although overall budget overrun was \$23,239.65, this was largely due to overspending on milestone completions, with a variance of -6.12%.
	Emmanuel	Confirmed that cost savings in project management tasks, particularly by reallocating tasks to operations specialists, helped balance out inefficiencies early on.
Overallocation Incident		
	Flippie	The overallocation incident in Period 6 caused a task delay that extended the project timeline from 13 February to 20 February 2026. This was partially recovered in later rounds, finishing only 7 days late.
	Lauren	- Noted this as a critical learning point on the importance of proactively managing resource loading. - These insights were recorded and helped a great deal in the informing allocations for later periods.
Resource Strategy		



	Lauren	- Presented a detailed account of resource strategy: - Senior resources were allocated to critical path tasks. - Underperforming or idle resources were phased out to reduce costs. - Heavy early-stage investment in training (Marthe, Kong, Valene) aimed at boosting long-term efficiency. - Team-building events (pizza parties, family event) were introduced in rounds 5 and 9 to boost morale.
	Flippie	Agreed that this iterative approach to resource development contributed directly to the jump from 8th to 1st place by Round 8 in the simulation.
Risk Management Adaptation		
	Lauren	- Reflected on risk mitigation measures, such as the decision to fire the Project Manager mid-simulation and replace with trained engineers and operations specialists. Also noted the success of contingency planning and adaptation to real-time simulation changes. - Many risks (especially risks other than managerial risks, which was the focus in the earlier risk register) were added throughout the second half of the project.

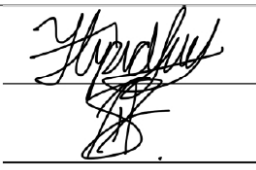
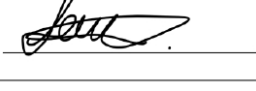


	Taro	Mentioned the importance of continuous risk reassessment as unforeseen events such as budget cuts and additional budget added in periods 5 and 8.
Comparing Real and Simulated Projects		
	Lauren	Urged everyone to reflect on how lessons from the simulation aligned with the real-life project.
	Taro	Observed that both projects emphasised the need for flexible planning where initial assumptions often needed to be adapted mid-execution.
	Emmanuel	Highlighted the value of real-time data tracking (via Sim4Projects) and its relevance to real-world metrics monitoring.
	Flippie	Pointed out that the coordination across different software platforms (MSProject, Sim4Projects, Excel) mirrored the integration challenges in Takealot's actual warehouse systems.
Multidisciplinary Collaboration		
	Emmanuel	Spoke on how working with mechanical and industrial engineers expanded his understanding of system-level design constraints and operational logistics.
	Luke	Commented on how exposure to data-driven decision-making (from data engineers) contributed clarity



		surrounding the budget forecasting process.
	Flippie	Noted the value in seeing mechanical components through a mechatronic lens, particularly when integrating robotic platforms. Attention was also an important skill brought to the table.
	Taro	Emphasised how the disciplines are very diverse and focused but are all still aligned in their analytical thinking style. This opens the door for very efficient collaboration.
Lessons Learned		
	Group Consensus	- Always plan for resource constraints early—acquire critical team members ahead of peak need. - Avoid overreliance on junior staff for key deliverables despite cost appeal. - Monitor team morale and cohesion actively—morale boosts correlate with performance gains. - Adopt rolling evaluations of performance metrics, not just static milestones.
Conclusion		
	Group Consensus	- The entire group reiterated that the project had been a success on both simulated and real-world fronts, and offered interesting insights into software integration, team strategy, and adaptive planning. - The team is happy with internal relations after this project and commends each member for contributing their best.



		Everyone is excited and proud that we won our Sim4Projects group competition.
Member Signatures		
	Flippie van der Spuy	    
	Taro Samuelson	
	Emmanuel Fakó	
	Luke Neethling	
	Lauren Coetser	
Adjournment		20:35

Addendums

- Assignment 3 Report
- Assignment 3 EVA Excel Document
- All previous meeting minutes

7 References

Steyn, J. M. N. a. H., 2021. - *Project management for engineering, business and technology (2021).pdf*. 6th ed. New York: Routledge.